

Question 1:

What is the primary use case of Apache Kafka?

- A) Real-time stream processing
- B) Data storage and retrieval
- C) Real-time data visualization
- D) High-performance machine learning

Answer: B) Data storage and retrieval

Question 2:

Which of the following is NOT a key feature of Apache Spark?

- A) Fault tolerance
- B) Real-time stream processing
- C) In-memory processing
- D) Distributed computing

Answer: B) Real-time stream processing

Question 3:

In Apache Kafka, what is a consumer group?

- A) A group of consumers sharing the same group ID to read data from different topics simultaneously.
- B) A group of consumers that have a shared state and consume messages from the same topic in parallel.
- C) A group of brokers responsible for handling consumer requests.
- D) A group of producers that produce messages to the same topic.

Answer: A) A group of consumers sharing the same group ID to read data from different topics simultaneously.

Question 4:

Which of the following is true regarding Spark's RDD (Resilient Distributed Dataset)?

- A) RDD is immutable and can only be modified through transformations.
- B) RDD is a mutable data structure that can be modified in place.
- C) RDD is an acronym for Real-time Data Distribution.
- D) RDD is a specialized data storage format used by Spark for efficient data retrieval.

Answer: A) RDD is immutable and can only be modified through transformations.

Question 5:

In Kafka, what is the role of a broker?

- A) Brokers are responsible for processing data streams in real-time.
- B) Brokers are responsible for storing and managing the messages in topics.
- C) Brokers are the consumers of data from producers.

D) Brokers are responsible for optimizing Spark jobs.

Answer: B) Brokers are responsible for storing and managing the messages in topics.

Question 6:

What does Spark's lazy evaluation mean?

- A) Spark doesn't start processing data until an action is triggered.
- B) Spark processes data in real-time without any delay.
- C) Spark processes data in batches, waiting for a specific time interval.
- D) Spark evaluates data lazily and skips unnecessary computations.

Answer: A) Spark doesn't start processing data until an action is triggered.

Question 7:

Which of the following is true regarding Kafka topics?

- A) Topics are exclusive to a single consumer group.
- B) Topics can only be read sequentially by consumers.
- C) Topics are immutable and cannot be modified once created.
- D) Topics can only be created by the Kafka administrator.

Answer: C) Topics are immutable and cannot be modified once created.

Question 8:

What is the main advantage of using Kafka over traditional messaging systems?

- A) Kafka guarantees exactly-once message delivery semantics.
- B) Kafka provides a more straightforward API for data processing.
- C) Kafka supports real-time analytics and machine learning.
- D) Kafka has lower latency and higher throughput for data ingestion.

Answer: D) Kafka has lower latency and higher throughput for data ingestion.

Question 9:

Which of the following is a real-time stream processing framework often used with Apache Kafka?

- A) Apache Hadoop
- B) Apache Flink
- C) Apache Hive
- D) Apache Zeppelin

Answer: B) Apache Flink

Question 10:

In Kafka, what happens if a consumer lags behind the producer significantly?

- A) Kafka automatically discards the lagging consumer and starts a new one.
- B) Kafka stops the producer until the consumer catches up.
- C) Kafka automatically replays the messages for the lagging consumer.
- D) Kafka increases the message retention time to allow the consumer to catch up.

Answer: D) Kafka increases the message retention time to allow the consumer to catch up.

Question 11:

What is the primary programming language used for writing Apache Spark applications?

- A) Java
- B) Python
- C) C++
- D) Scala

Answer: D) Scala

Question 12:

Which of the following is NOT a Spark deployment mode?

- A) Standalone Mode
- B) YARN Mode
- C) Kubernetes Mode
- D) Cluster Mode

Answer: C) Kubernetes Mode

Question 13:

In Kafka, what is the purpose of a producer's acknowledgment setting "all"?

- A) The producer acknowledges messages only when all consumers have processed them.
- B) The producer waits for acknowledgment from all replicas before considering a message as sent.
- C) The producer acknowledges messages only when they have been processed by all partitions.
- D) The producer waits for acknowledgment from all consumers before sending the next message.

Answer: B) The producer waits for acknowledgment from all replicas before considering a message as sent.

Question 14:

Which Spark component is responsible for scheduling tasks within the cluster?

- A) Driver
- B) Executor
- C) Master
- D) Scheduler

Answer: D) Scheduler

Question 15:

What is the main difference between Kafka and traditional message queuing systems like RabbitMQ or ActiveMQ?

- A) Kafka supports distributed stream processing, while traditional message queuing systems do not.
- B) Kafka provides higher throughput and lower latency, while traditional systems have lower performance.
- C) Kafka uses a push-based model, while traditional systems use a pull-based model for message consumption.
- D) Kafka guarantees message delivery in the order they are produced, while traditional systems may not.

Answer: C) Kafka uses a push-based model, while traditional systems use a pull-based model for message consumption.

Question 16:

Which of the following is a benefit of using Spark's DataFrame API over RDDs?

- A) DataFrame API provides more fine-grained control over partitioning and caching.
- B) DataFrame API allows for lazy evaluation and optimized query optimization.
- C) DataFrame API is more memory-efficient and requires less garbage collection.
- D) DataFrame API provides a more expressive and user-friendly syntax for data processing.

Answer: D) DataFrame API provides a more expressive and user-friendly syntax for data processing.

Question 17:

What is the role of Apache ZooKeeper in Apache Kafka?

- A) ZooKeeper is used for distributed locking and coordination of Kafka brokers.
- B) ZooKeeper is responsible for storing and managing the data in Kafka topics.
- C) ZooKeeper is the primary messaging system used in Kafka's data pipeline.
- D) ZooKeeper provides real-time analytics and data visualization capabilities.

Answer: A) ZooKeeper is used for distributed locking and coordination of Kafka brokers.

Question 18:

Which Spark deployment mode allows multiple Spark applications to share the same cluster?

- A) Standalone Mode
- B) YARN Mode
- C) Kubernetes Mode
- D) Cluster Mode

Answer: B) YARN Mode

Question 19:

What is the primary advantage of using Spark Streaming over traditional batch processing systems like Apache Hadoop MapReduce?

- A) Spark Streaming guarantees exactly-once processing semantics.
- B) Spark Streaming provides lower latency for real-time data processing.
- C) Spark Streaming allows for simpler and more concise data processing code.
- D) Spark Streaming supports data processing across multiple nodes.

Answer: B) Spark Streaming provides lower latency for real-time data processing.

Question 20:

Which of the following is true about Spark Structured Streaming?

- A) Spark Structured Streaming is based on RDDs (Resilient Distributed Datasets).
- B) Spark Structured Streaming is a separate project and not integrated with Spark.
- C) Spark Structured Streaming processes data in micro-batches rather than traditional RDD batches.
- D) Spark Structured Streaming is primarily used for batch processing, not real-time processing.

Answer: C) Spark Structured Streaming processes data in micro-batches rather than traditional RDD batches.

Question 1:

Which of the following is NOT a component of Apache Spark?

- A) Spark Core
- B) Spark SQL
- C) Spark Streaming
- D) Spark Queue

Answer: D) Spark Queue

Question 2:

In Apache Kafka, what is the role of a ZooKeeper?

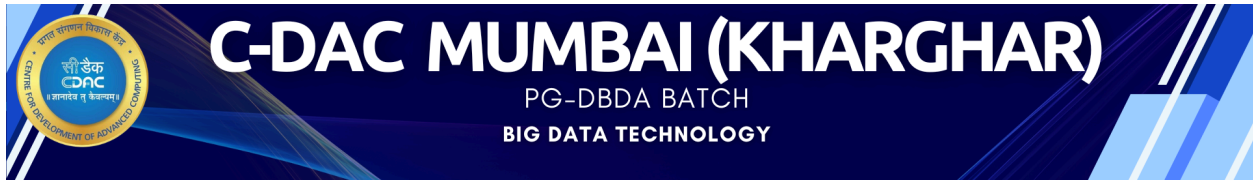
- A) ZooKeeper is used for managing the metadata of Kafka topics.
- B) ZooKeeper is responsible for data processing in Kafka.
- C) ZooKeeper provides real-time analytics for Kafka consumers.
- D) ZooKeeper is used for distributed coordination and leadership election in Kafka.

Answer: D) ZooKeeper is used for distributed coordination and leadership election in Kafka.

Question 3:

What does Kafka use to ensure fault tolerance and replication of messages?

- A) Quorum Replication
- B) Leader-Follower Model



- C) Multi-Datcenter Replication
- D) Paxos Consensus Algorithm

Answer: B) Leader-Follower Model

Question 4:

Which Spark transformation allows you to perform an operation on two separate RDDs and return a new RDD containing the result?

- A) union()
- B) join()
- C) intersect()
- D) subtract()

Answer: B) join()

Question 5:

What is the primary use case of Apache Kafka?

- A) Real-time stream processing
- B) Data storage and retrieval
- C) High-performance machine learning
- D) Real-time data visualization

Answer: B) Data storage and retrieval

Question 6:

Which of the following is true regarding Spark's lineage graph?

- A) It represents the parallel execution plan of a Spark job.
- B) It tracks the dependencies between RDDs in a Spark application.
- C) It is used to visualize the real-time data flow in a Spark cluster.
- D) It stores the historical data lineage of RDDs.

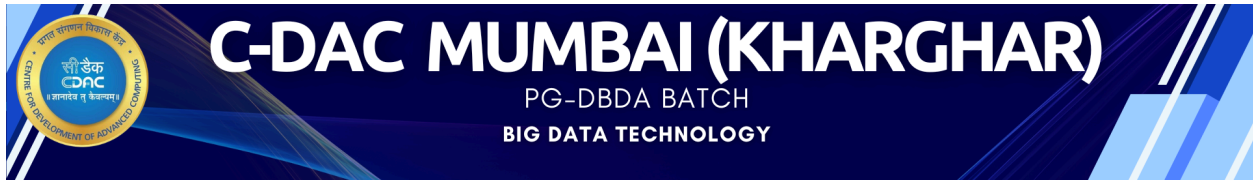
Answer: B) It tracks the dependencies between RDDs in a Spark application.

Question 7:

In Kafka, what happens if a consumer fails to send heartbeats to the group coordinator?

- A) The consumer is automatically removed from the consumer group.
- B) The consumer's offset is reset to the beginning of the topic.
- C) The consumer's offset is reset to the last committed offset.
- D) The consumer is flagged for a manual reassignment of partitions.

Answer: A) The consumer is automatically removed from the consumer group.



Question 8:

Which of the following is NOT a property of Apache Kafka?

- A) Scalability
- B) Fault tolerance
- C) Real-time stream processing
- D) Exactly-once message delivery

Answer: C) Real-time stream processing

Question 9:

What is the benefit of using Spark's Broadcast variable?

- A) Broadcast variables allow sharing data efficiently across tasks in a Spark job.
- B) Broadcast variables enable real-time data visualization in Spark applications.
- C) Broadcast variables reduce the memory usage of Spark executors.
- D) Broadcast variables automatically handle fault tolerance in Spark.

Answer: A) Broadcast variables allow sharing data efficiently across tasks in a Spark job.

Question 10:

In Kafka, what is the purpose of a consumer offset?

- A) It indicates the size of the Kafka topic.
- B) It tracks the progress of a consumer in a topic.
- C) It represents the unique ID of a consumer in a group.
- D) It denotes the number of partitions in a Kafka cluster.

Answer: B) It tracks the progress of a consumer in a topic.

Question 11:

Which of the following Spark actions triggers the execution of transformations and returns results to the driver program?

- A) collect()
- B) count()
- C) reduce()
- D) save()

Answer: A) collect()

Question 12:

What is the role of Apache ZooKeeper in Apache Kafka?

- A) ZooKeeper provides real-time analytics and data visualization capabilities.
- B) ZooKeeper is responsible for data storage and retrieval in Kafka.
- C) ZooKeeper is used for distributed locking and coordination of Kafka brokers.

D) ZooKeeper acts as a consumer in Kafka, processing data from producers.

Answer: C) ZooKeeper is used for distributed locking and coordination of Kafka brokers.

Question 13:

Which Spark deployment mode is suitable for local development and testing on a single machine?

- A) Standalone Mode
- B) YARN Mode
- C) Kubernetes Mode
- D) Cluster Mode

Answer: A) Standalone Mode

Question 14:

What is the significance of Kafka partitions in a topic?

- A) Partitions are used to distribute Kafka brokers in a cluster.
- B) Partitions are used to manage consumer groups in Kafka.
- C) Partitions enable parallelism and scalability in Kafka.
- D) Partitions store the metadata of Kafka topics.

Answer: C) Partitions enable parallelism and scalability in Kafka.

Question 15:

What does Apache Spark's DAG (Directed Acyclic Graph) represent?

- A) It represents the execution plan of Spark transformations.
- B) It visualizes the data flow between Spark RDDs.
- C) It is used for partitioning data in Spark applications.
- D) It stores the historical lineage of Spark jobs.

Answer: A) It represents the execution plan of Spark transformations.

Question 16:

Which of the following is a benefit of using Kafka Connect?

- A) Kafka Connect allows real-time stream processing with Apache Spark.
- B) Kafka Connect enables seamless data integration between Kafka and Spark.
- C) Kafka Connect automatically handles consumer offset management.
- D) Kafka Connect is a messaging system used for collecting and storing data.

Answer: B) Kafka Connect enables seamless data integration between Kafka and Spark.

Question 17:

What is the primary programming language used for writing Apache Kafka applications?



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- A) Java
- B) Python
- C) C++
- D) Scala

Answer: A) Java

Question 18:

In Spark Streaming, what is the role of a micro-batch?

- A) Micro-batch represents the time interval for batch processing in Spark.
- B) Micro-batch is a small portion of data processed in real-time.
- C) Micro-batch is a parallel computing unit in a Spark cluster.
- D) Micro-batch is the smallest unit of data in Spark Streaming.

Answer: B) Micro-batch is a small portion of data processed in real-time.

Question 19:

Which of the following Spark transformations allows you to filter elements based on a condition and return a new RDD?

- A) map()
- B) reduce()
- C) filter()
- D) groupBy()

Answer: C) filter()

Question 20:

What is the primary advantage of using Kafka's log compaction feature?

- A) Log compaction reduces the size of Kafka topics, saving storage space.
- B) Log compaction provides real-time analytics and data visualization capabilities.
- C) Log compaction guarantees exactly-once message delivery semantics.
- D) Log compaction ensures that Kafka retains the latest value of each key in a topic.

Answer: D) Log compaction ensures that Kafka retains the latest value of each key in a topic.

Question 1:

What is the main advantage of using Apache Kafka over traditional message brokers like RabbitMQ or ActiveMQ?

- A) Kafka supports exactly-once message delivery semantics.
- B) Kafka provides real-time stream processing capabilities.
- C) Kafka allows seamless integration with Apache Spark.
- D) Kafka offers lower latency and higher throughput for data ingestion.

Answer: D) Kafka offers lower latency and higher throughput for data ingestion.

Question 2:

Which of the following is NOT a core abstraction in Apache Spark?

- A) RDD (Resilient Distributed Dataset)
- B) DataFrame
- C) DStream (Discretized Stream)
- D) Dataset

Answer: C) DStream (Discretized Stream)

Question 3:

What is the primary use case of Spark Structured Streaming?

- A) Real-time stream processing
- B) Batch processing
- C) Interactive data analysis
- D) Machine learning

Answer: A) Real-time stream processing

Question 4:

In Kafka, which component is responsible for persisting the messages on disk?

- A) Producers
- B) Brokers
- C) Consumers
- D) ZooKeeper

Answer: B) Brokers

Question 5:

What does Spark's lineage information represent?

- A) It tracks the data lineage of RDDs for fault tolerance.
- B) It stores the schema information of DataFrames.
- C) It represents the parallel execution plan of Spark jobs.
- D) It keeps a log of Spark streaming events.

Answer: A) It tracks the data lineage of RDDs for fault tolerance.

Question 6:

Which of the following is NOT true about Kafka topics?

- A) Topics can be partitioned to achieve parallelism.

- B) Topics are stored on the Kafka brokers.
- C) A topic can have only one consumer group.
- D) Producers publish messages to topics.

Answer: C) A topic can have only one consumer group.

Question 7:

What is the significance of Spark's Catalyst optimizer?

- A) It optimizes Kafka consumer group coordination for parallel processing.
- B) It optimizes the execution plan of Spark SQL queries.
- C) It ensures exactly-once message delivery semantics in Kafka.
- D) It optimizes data storage in Spark RDDs.

Answer: B) It optimizes the execution plan of Spark SQL queries.

Question 8:

Which of the following is true about Apache Spark's streaming capabilities?

- A) Spark Streaming processes data in real-time without any delay.
- B) Spark Streaming supports low-level, micro-batch processing only.
- C) Spark Streaming guarantees exactly-once processing semantics.
- D) Spark Streaming can process data from Kafka topics directly without any transformation.

Answer: B) Spark Streaming supports low-level, micro-batch processing only.

Question 9:

In Kafka, how does the partitioning of data in a topic affect the order of messages?

- A) Messages in the same partition are always processed in order, but the order is not guaranteed across different partitions.
- B) Kafka guarantees that messages are always processed in the order they are produced, regardless of partitioning.
- C) Kafka ensures that messages are processed in a round-robin manner across all partitions for load balancing.
- D) Messages from different partitions are processed in parallel, but the order is not guaranteed even within the same partition.

Answer: A) Messages in the same partition are always processed in order, but the order is not guaranteed across different partitions.

Question 10:

What is the role of a Spark Driver in a Spark application?

- A) The Driver is responsible for processing data in real-time and generating streaming results.
- B) The Driver represents the Kafka consumer group coordinator.

- C) The Driver runs on each executor node to coordinate data processing tasks.
- D) The Driver runs on the master node and maintains the SparkContext.

Answer: D) The Driver runs on the master node and maintains the SparkContext.

Question 11:

Which of the following is true about Apache Kafka's message retention policy?

- A) Kafka retains messages forever by default to ensure data durability.
- B) Kafka uses a FIFO (First-In-First-Out) approach and discards old messages after a certain time.
- C) Kafka's message retention policy cannot be configured; it is fixed at 7 days.
- D) Kafka's message retention policy is determined by the number of consumers in a consumer group.

Answer: B) Kafka uses a FIFO (First-In-First-Out) approach and discards old messages after a certain time.

Question 12:

In Apache Spark, which transformation should be used to remove duplicates from an RDD or DataFrame?

- A) distinct()
- B) filter()
- C) dropDuplicates()
- D) removeDuplicates()

Answer: C) dropDuplicates()

Question 13:

Which of the following is a true statement about Spark DataFrame and Spark SQL?

- A) Spark DataFrame is an abstraction layer on top of Spark SQL.
- B) Spark SQL is a new programming language introduced in Apache Spark.
- C) Spark DataFrame is used for batch processing, while Spark SQL is used for stream processing.
- D) Spark DataFrame and Spark SQL are two different ways of referring to the same concept.

Answer: D) Spark DataFrame and Spark SQL are two different ways of referring to the same concept.

Question 14:

What is the role of a Kafka consumer group?

- A) Consumer groups ensure that messages are consumed in parallel for better performance.
- B) Consumer groups manage the distribution of messages across Kafka partitions.
- C) Consumer groups coordinate the work of Kafka brokers in a cluster.
- D) Consumer groups ensure that messages are processed only once by the consumers.

Answer: B) Consumer groups manage the distribution of messages across Kafka partitions.

Question 15:

In Spark Streaming, what does the "window" operation allow you to do?

- A) It allows you to define the time window for data processing in Spark.
- B) It enables you to join two DataFrames together based on a common key.
- C) It allows you to remove outdated data from an RDD.
- D) It enables you to aggregate data over a sliding or tumbling time window.

Answer: D) It enables you to aggregate data over a sliding or tumbling time window.

Question 16:

What is the purpose of Kafka Connect?

- A) Kafka Connect is used to connect Kafka with external databases.
- B) Kafka Connect is a Kafka consumer used for real-time data processing.
- C) Kafka Connect is a tool for managing Kafka consumer group coordination.
- D) Kafka Connect is used to integrate Kafka with external systems for data import and export.

Answer: D) Kafka Connect is used to integrate Kafka with external systems for data import and export.

Question 17:

Which of the following is NOT true about Apache Spark's Broadcast variable?

- A) Broadcast variables are used to share read-only data efficiently across tasks in a Spark job.
- B) Broadcast variables are useful when a large dataset needs to be shared across all nodes in a cluster.
- C) Broadcast variables can be modified in place during the execution of a Spark job.
- D) Broadcast variables help reduce the amount of data that needs to be sent over the network during data processing.

Answer: C) Broadcast variables can be modified in place during the execution of a Spark job.

Question 18:

What is the primary benefit of using Apache Spark's Structured Streaming over traditional Spark Streaming?

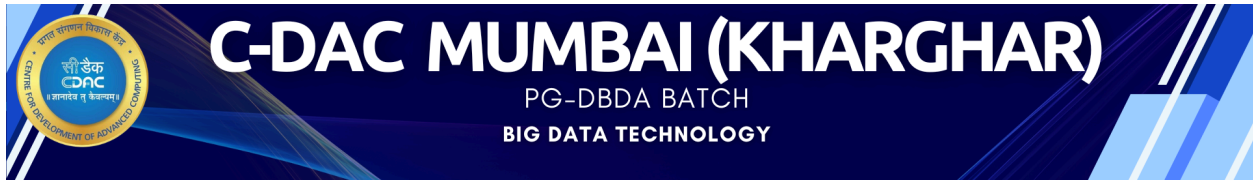
- A) Structured Streaming supports real-time processing of unstructured data.
- B) Structured Streaming guarantees exactly-once message delivery semantics.
- C) Structured Streaming provides higher throughput and lower latency for data processing.
- D) Structured Streaming allows developers to use the DataFrame API for stream processing.

Answer: D) Structured Streaming allows developers to use the DataFrame API for stream processing.

Question 19:

What is the significance of the Kafka Consumer Offset?

- A) The Consumer Offset represents the size of the consumer group in a Kafka topic.
- B) The Consumer Offset is a unique ID assigned to each message in a Kafka topic.



- C) The Consumer Offset represents the current position of a consumer within a Kafka topic.
- D) The Consumer Offset is the Kafka broker responsible for handling consumer requests.

Answer: C) The Consumer Offset represents the current position of a consumer within a Kafka topic.

Question 20:

In Spark, how can you achieve fault tolerance when processing data using the DataFrame API?

- A) By using the "checkpoint" operation after each transformation.
- B) By setting the "failOnDataLoss" parameter to "true" in the SparkContext configuration.
- C) By enabling the "DataFrame Persistence" option in the Spark UI.
- D) By using the "cache" operation to persist intermediate results in memory.

Answer: D) By using the "cache" operation to persist intermediate results in memory.

Question: Which Spark API is primarily used for interactive querying and processing structured data?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: b) SparkSQL

Question: Which API in Spark provides machine learning functionalities and algorithms?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

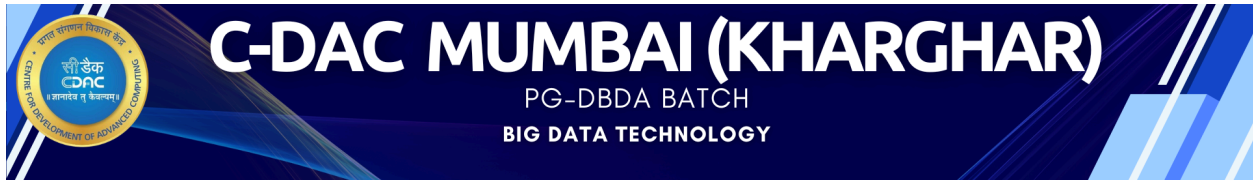
Answer: d) SparkMLlib

Question: Which Spark API allows you to process streaming data in real-time?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: c) SparkStreaming

Question: What is the primary purpose of Spark Core API?



- a) Data visualization
- b) Machine learning
- c) Distributed data processing
- d) Real-time streaming

Answer: c) Distributed data processing

Question: In Spark, what does RDD stand for?

- a) Relational Data Definitions
- b) Resilient Distributed Datasets
- c) Real-time Data Driven
- d) Rapid Data Distribution

Answer: b) Resilient Distributed Datasets

Question: Which of the following Spark APIs provides higher-level abstractions for working with structured data?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: b) SparkSQL

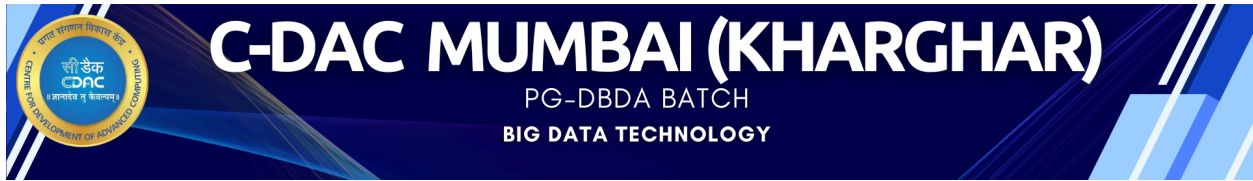
Question: Which Spark API supports writing and executing SQL-like queries on distributed datasets?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: b) SparkSQL

Question: Which Spark API is used for building distributed data processing applications?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib



Answer: a) SparkCore

Question: Which Spark API is used for machine learning tasks?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: d) SparkMLlib

Question: What is the significance of lineage information in Spark?

- a) It helps in data visualization.
- b) It enables fault tolerance and data recovery.
- c) It optimizes machine learning algorithms.
- d) It improves real-time streaming performance.

Answer: b) It enables fault tolerance and data recovery.

Question: In Spark, what does DAG stand for?

- a) Distributed Application Generator
- b) Directed Acyclic Graph
- c) Data Access Gateway
- d) Dynamic Algorithm Generator

Answer: b) Directed Acyclic Graph

Question: Which API is responsible for high-level abstraction and optimization of Spark jobs?

- a) SparkCore
- b) SparkSQL
- c) Catalyst
- d) SparkMLlib

Answer: c) Catalyst

Question: Which Spark API allows the execution of batch processing jobs?

- a) SparkCore
- b) SparkSQL
- c) SparkBatch
- d) SparkMLlib

Answer: a) SparkCore

Question: Which Spark API is used for working with graph data?

- a) SparkCore
- b) GraphX
- c) SparkStreaming
- d) SparkMLlib

Answer: b) GraphX

Question: In Spark, what is the main role of the "Driver" program?

- a) It manages the Spark UI for job monitoring.
- b) It coordinates tasks and schedules execution.
- c) It handles data visualization.
- d) It performs data cleaning operations.

Answer: b) It coordinates tasks and schedules execution.

Question: Which Spark API provides built-in support for processing XML and JSON data formats?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: b) SparkSQL

Question: Which Spark API is used for real-time data processing from sources like Kafka and Flume?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: c) SparkStreaming

Question: Which Spark API is suitable for iterative machine learning algorithms?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: d) SparkMLlib

Question: What is the primary advantage of using DataFrame over RDD in Spark?

- a) DataFrame offers better fault tolerance.
- b) DataFrame supports distributed machine learning.
- c) DataFrame provides better performance optimization.
- d) DataFrame allows real-time streaming.

Answer: c) DataFrame provides better performance optimization.

Question: Which API is responsible for managing memory and data storage in Spark?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) Tungsten

Answer: d) Tungsten

Question: Which Spark API is designed for interactive querying and processing semi-structured data?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: b) SparkSQL

Question: In Spark, what is the role of the "Executor" in the cluster?

- a) It processes Spark jobs on the driver node.
- b) It manages the data storage on HDFS.
- c) It monitors and controls the Spark applications.
- d) It performs computations and stores data in memory or disk.

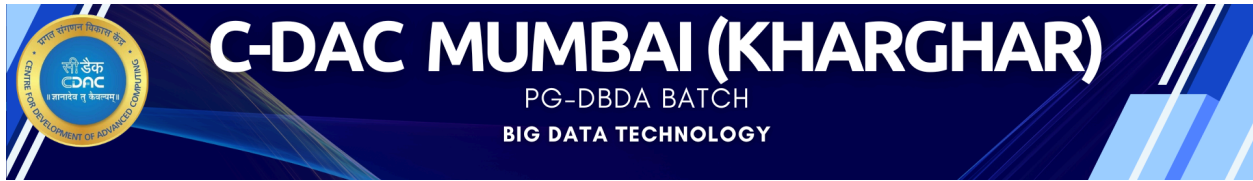
Answer: d) It performs computations and stores data in memory or disk.

Question: Which Spark API allows you to schedule jobs based on time intervals?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkScheduler

Answer: c) SparkStreaming

Question: Which API provides higher-level abstractions for machine learning pipelines in Spark?



- a) SparkCore
- b) SparkSQL
- c) MLlib
- d) SparkMLlib

Answer: d) SparkMLlib

Question: Which Spark API provides the functionality of running SQL-like queries on large-scale data?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: b) SparkSQL

Question: What is the role of the "Master" in a Spark cluster?

- a) It executes tasks and computations.
- b) It manages resources and schedules tasks.
- c) It handles data visualization.
- d) It optimizes machine learning algorithms.

Answer: b) It manages resources and schedules tasks.

Question: Which Spark API provides support for working with structured data and integrates with external data sources?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: b) SparkSQL

Question: Which API in Spark is responsible for managing data partitioning and distribution across the cluster?

- a) SparkCore
- b) SparkSQL
- c) Catalyst
- d) SparkMLlib

Answer: a) SparkCore

Question: Which Spark API provides support for distributed graph processing?

- a) SparkCore
- b) GraphX
- c) SparkStreaming
- d) SparkMLlib

Answer: b) GraphX

Question: What is the primary role of the "Cluster Manager" in Spark?

- a) It handles data replication across the cluster.
- b) It optimizes the Spark job execution plan.
- c) It manages resources and schedules Spark applications.
- d) It performs data cleaning operations.

Answer: c) It manages resources and schedules Spark applications.

Question: Which Spark API provides support for stream processing?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: c) SparkStreaming

Question: Which API in Spark is responsible for managing task execution and fault tolerance?

- a) SparkCore
- b) SparkSQL
- c) SparkScheduler
- d) SparkMLlib

Answer: c) SparkScheduler

Question: In Spark, what is the role of the "ApplicationMaster" in a YARN cluster?

- a) It manages data storage and caching.
- b) It coordinates and negotiates resources with YARN ResourceManager.
- c) It performs data visualization.
- d) It optimizes machine learning algorithms.

Answer: b) It coordinates and negotiates resources with YARN ResourceManager.

Question: Which Spark API is responsible for handling Spark job monitoring and providing a web-based user interface?

- a) SparkUI

- b) SparkCore
- c) SparkStreaming
- d) SparkMLlib

Answer: a) SparkUI

Question: In Spark, what is the role of the "Application Driver"?

- a) It coordinates tasks within a stage.
- b) It optimizes the Spark job execution plan.
- c) It monitors the Spark applications.
- d) It processes Spark jobs on the driver node.

Answer: c) It monitors the Spark applications.

Question: Which Spark API is used for ETL (Extract, Transform, Load) operations?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: b) SparkSQL

Question: Which Spark API provides built-in support for working with text files and data formats like CSV and JSON?

- a) SparkCore
- b) SparkSQL
- c) SparkFileIO
- d) SparkMLlib

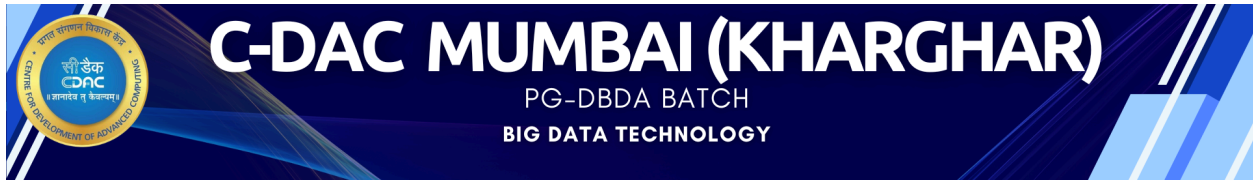
Answer: a) SparkCore

Question: What is the primary advantage of using Spark's DataFrame API over traditional RDDs for SQL-like operations?

- a) Spark DataFrame provides better fault tolerance.
- b) Spark DataFrame allows real-time streaming.
- c) Spark DataFrame optimizes data processing performance.
- d) Spark DataFrame provides support for graph processing.

Answer: c) Spark DataFrame optimizes data processing performance.

Question: Which Spark API provides support for distributed machine learning algorithms and model training?



- a) SparkCore
- b) SparkSQL
- c) MLlib
- d) SparkMLlib

Answer: d) SparkMLlib

Question: In Spark, what is the significance of the "shuffle" operation?

- a) It sorts the data in descending order.
- b) It performs a random reordering of data.
- c) It combines and redistributes data across partitions.
- d) It groups data based on a given key.

Answer: c) It combines and redistributes data across partitions.

Question: Which Spark API is designed for processing data streams in micro-batches?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: c) SparkStreaming

Question: Which Spark API allows you to process data in parallel across a cluster of machines?

- a) SparkCore
- b) SparkSQL
- c) SparkParallel
- d) SparkMLlib

Answer: a) SparkCore

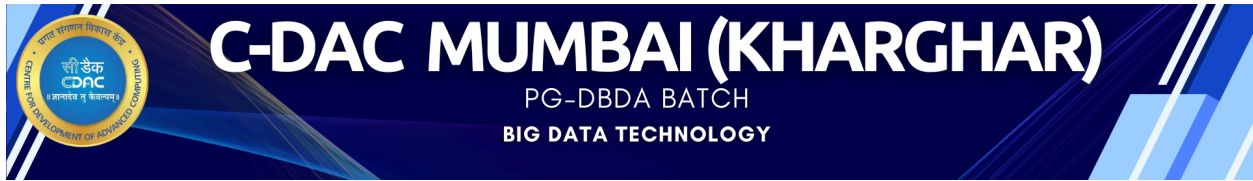
Question: In Spark, what is the role of the "BlockManager"?

- a) It handles data replication across the cluster.
- b) It manages resources and schedules Spark applications.
- c) It performs data cleaning operations.
- d) It manages data storage and caching.

Answer: d) It manages data storage and caching.

Question: Which Spark API is used for processing structured and semi-structured data from various data sources?

- a) SparkCore



- b) SparkSQL
- c) SparkDataIO
- d) SparkMLlib

Answer: b) SparkSQL

Question: What is the primary advantage of using Spark Streaming over traditional batch processing?

- a) Spark Streaming provides better fault tolerance.
- b) Spark Streaming supports real-time data processing.
- c) Spark Streaming optimizes machine learning algorithms.
- d) Spark Streaming handles data visualization.

Answer: b) Spark Streaming supports real-time data processing.

Question: Which Spark API provides support for distributed deep learning?

- a) SparkCore
- b) SparkSQL
- c) SparkDeep
- d) SparkMLlib

Answer: d) SparkMLlib

Question: In Spark, which component is responsible for managing the SparkContext?

- a) SparkCore
- b) SparkContext
- c) SparkDriver
- d) SparkMaster

Answer: b) SparkContext

Question: Which Spark API is used for working with time-series data?

- a) SparkCore
- b) SparkSQL
- c) SparkTimeSeries
- d) SparkMLlib

Answer: b) SparkSQL

Question: In Spark, what is the role of the "Worker" in the cluster?

- a) It processes Spark jobs on the driver node.
- b) It manages the data storage on HDFS.
- c) It performs computations and stores data in memory or disk.

d) It monitors and controls the Spark applications.

Answer: c) It performs computations and stores data in memory or disk.

Question: Which Spark API is used for handling complex data processing tasks that require low-level control?

- a) SparkCore
- b) SparkSQL
- c) SparkAdvanced
- d) SparkMLlib

Answer: a) SparkCore

Question: Which Apache Spark API is designed specifically for real-time data streaming and processing?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: c) SparkStreaming

Question: In Apache Spark, which API is the foundation and core of the entire framework?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

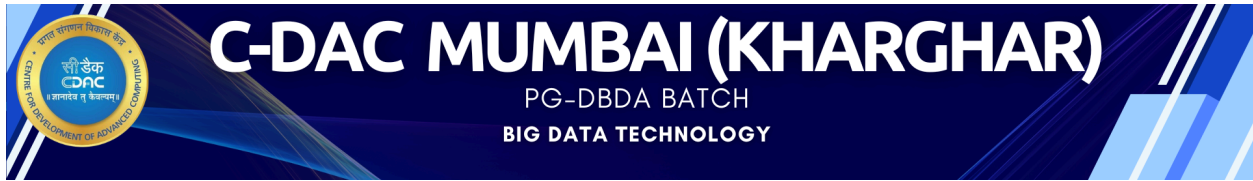
Answer: a) SparkCore

Question: Which Spark API provides support for machine learning and advanced analytics?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: d) SparkMLlib

Question: What does RDD stand for in Apache Spark?



- a) Resilient Data Division
- b) Reactive Distributed Data
- c) Resilient Distributed Datasets
- d) Real-time Data Dispatch

Answer: c) Resilient Distributed Datasets

Question: Which Spark API is primarily used for processing structured data using SQL queries?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: b) SparkSQL

Question: Which Apache Spark API provides support for graph processing?

- a) SparkCore
- b) SparkSQL
- c) GraphX
- d) SparkMLlib

Answer: c) GraphX

Question: In Apache Spark, which component is responsible for distributing tasks across the cluster?

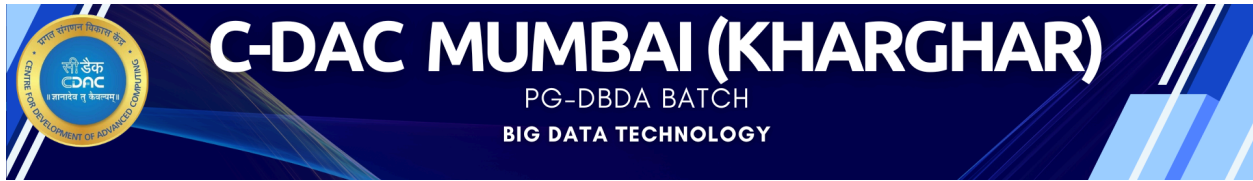
- a) SparkContext
- b) SparkScheduler
- c) SparkMaster
- d) SparkDriver

Answer: b) SparkScheduler

Question: Which Spark API is responsible for memory management and optimization?

- a) SparkCore
- b) SparkSQL
- c) SparkMemory
- d) Tungsten

Answer: d) Tungsten



Question: Which Spark API allows you to process data in real-time from sources like Kafka and Flume?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: c) SparkStreaming

Question: In Spark, what is the role of the "Driver" program?

- a) It processes Spark jobs on the driver node.
- b) It coordinates tasks and schedules execution.
- c) It manages resources and schedules Spark applications.
- d) It performs data cleaning operations.

Answer: b) It coordinates tasks and schedules execution.

Question: Which Apache Spark API provides built-in support for SQL queries on distributed datasets?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: b) SparkSQL

Question: Which Spark API is suitable for iterative machine learning algorithms?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: d) SparkMLlib

Question: In Spark, what does DAG stand for?

- a) Data Analysis Graph
- b) Directed Acyclic Graph
- c) Distributed Analytics Group
- d) Data Aggregation Grid

Answer: b) Directed Acyclic Graph

Question: Which Spark API provides support for distributed machine learning?

- a) SparkCore
- b) SparkSQL
- c) SparkMLlib
- d) SparkStreaming

Answer: c) SparkMLlib

Question: What is the primary purpose of Spark Core API?

- a) Real-time data processing
- b) Machine learning
- c) Data visualization
- d) Distributed data processing

Answer: d) Distributed data processing

Question: Which Spark API is used for interactive querying and processing structured data?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: b) SparkSQL

Question: Which Spark API provides support for working with semi-structured data like JSON and Avro?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

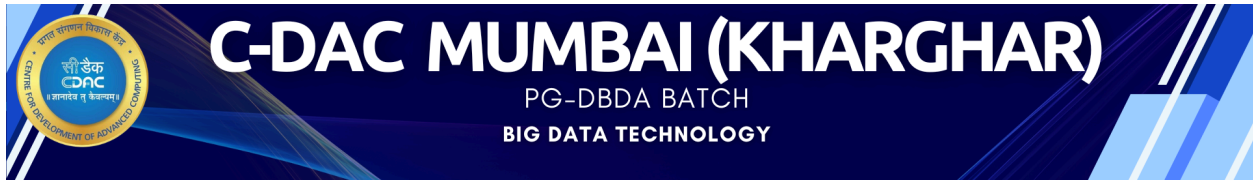
Answer: b) SparkSQL

Question: In Spark, what is the role of the "Executor" in the cluster?

- a) It processes Spark jobs on the driver node.
- b) It manages the data storage on HDFS.
- c) It monitors and controls the Spark applications.
- d) It performs computations and stores data in memory or disk.

Answer: d) It performs computations and stores data in memory or disk.

Question: Which Spark API allows the execution of batch processing jobs?



- a) SparkCore
- b) SparkSQL
- c) SparkBatch
- d) SparkMLlib

Answer: a) SparkCore

Question: What is the role of the "Master" in a Spark cluster?

- a) It executes tasks and computations.
- b) It manages resources and schedules tasks.
- c) It handles data visualization.
- d) It optimizes machine learning algorithms.

Answer: b) It manages resources and schedules tasks.

Question: Which Spark API provides higher-level abstractions for machine learning pipelines?

- a) SparkCore
- b) SparkSQL
- c) SparkMLlib
- d) SparkML

Answer: d) SparkML

Question: In Spark, what is the significance of lineage information?

- a) It helps in data visualization.
- b) It enables fault tolerance and data recovery.
- c) It optimizes machine learning algorithms.
- d) It improves real-time streaming performance.

Answer: b) It enables fault tolerance and data recovery.

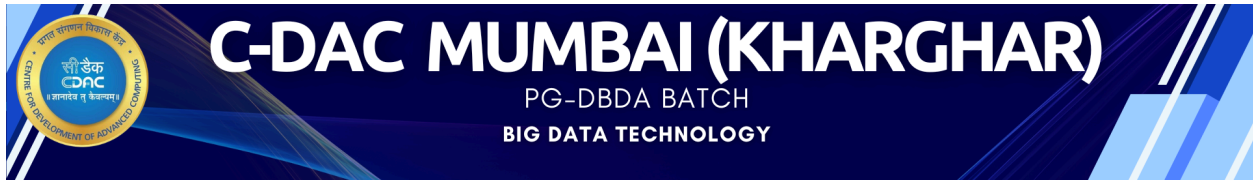
Question: Which Spark API is used for processing and analyzing large-scale time-series data?

- a) SparkCore
- b) SparkSQL
- c) SparkTS
- d) SparkMLlib

Answer: c) SparkTS

Question: Which Spark API provides support for distributed deep learning?

- a) SparkCore
- b) SparkSQL



- c) SparkDeep
- d) SparkMLlib

Answer: c) SparkDeep

Question: In Spark, what is the role of the "ApplicationMaster" in a YARN cluster?

- a) It manages data storage and caching.
- b) It coordinates and negotiates resources with YARN ResourceManager.
- c) It performs data visualization.
- d) It optimizes machine learning algorithms.

Answer: b) It coordinates and negotiates resources with YARN ResourceManager.

Question: Which Spark API is used for processing data in parallel across a cluster of machines?

- a) SparkCore
- b) SparkSQL
- c) SparkParallel
- d) SparkMLlib

Answer: a) SparkCore

Question: What is the primary advantage of using Spark's DataFrame API over traditional RDDs for SQL-like operations?

- a) Spark DataFrame provides better fault tolerance.
- b) Spark DataFrame allows real-time streaming.
- c) Spark DataFrame optimizes data processing performance.
- d) Spark DataFrame provides support for graph processing.

Answer: c) Spark DataFrame optimizes data processing performance.

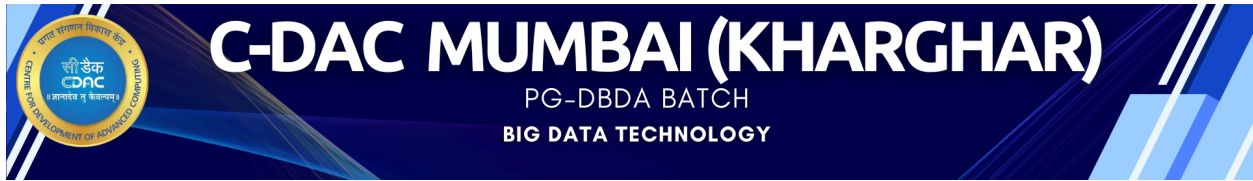
Question: Which Spark API is responsible for handling Spark job monitoring and providing a web-based user interface?

- a) SparkUI
- b) SparkCore
- c) SparkStreaming
- d) SparkMLlib

Answer: a) SparkUI

Question: Which Spark API is used for handling complex data processing tasks that require low-level control?

- a) SparkCore



- b) SparkSQL
- c) SparkAdvanced
- d) SparkMLlib

Answer: a) SparkCore

Question: In Spark, what is the role of the "Worker" in the cluster?

- a) It processes Spark jobs on the driver node.
- b) It manages the data storage on HDFS.
- c) It performs computations and stores data in memory or disk.
- d) It monitors and controls the Spark applications.

Answer: c) It performs computations and stores data in memory or disk.

Question: Which Spark API is used for working with time-series data?

- a) SparkCore
- b) SparkSQL
- c) SparkTimeSeries
- d) SparkMLlib

Answer: c) SparkTimeSeries

Question: Which Spark API is used for ETL (Extract, Transform, Load) operations?

- a) SparkCore
- b) SparkSQL
- c) SparkETL
- d) SparkMLlib

Answer: b) SparkSQL

Question: In Spark, what is the role of the "Application Driver"?

- a) It coordinates tasks within a stage.
- b) It optimizes the Spark job execution plan.
- c) It monitors the Spark applications.
- d) It processes Spark jobs on the driver node.

Answer: c) It monitors the Spark applications.

Question: Which Spark API is designed for processing data streams in micro-batches?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming

d) SparkMLlib

Answer: c) SparkStreaming

Question: Which Spark API is used for working with complex data types like graphs and matrices?

- a) SparkCore
- b) SparkSQL
- c) SparkGraph
- d) SparkMLlib

Answer: c) SparkGraph

Question: What is the primary advantage of using Spark Streaming over traditional batch processing?

- a) Spark Streaming provides better fault tolerance.
- b) Spark Streaming supports real-time data processing.
- c) Spark Streaming optimizes machine learning algorithms.
- d) Spark Streaming handles data visualization.

Answer: b) Spark Streaming supports real-time data processing.

Question: Which Spark API provides support for distributed graph processing?

- a) SparkCore
- b) SparkSQL
- c) SparkGraphX
- d) SparkMLlib

Answer: c) SparkGraphX

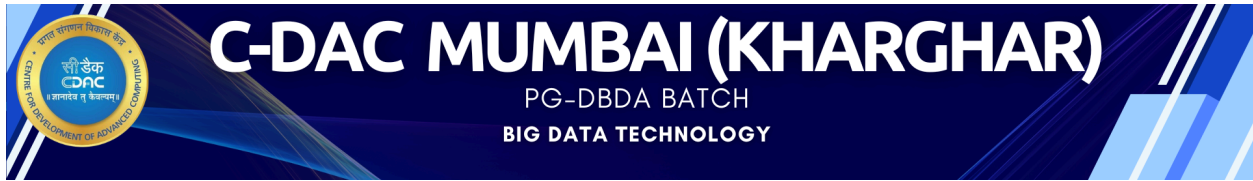
Question: Which Spark API is responsible for managing task execution and fault tolerance?

- a) SparkCore
- b) SparkSQL
- c) SparkScheduler
- d) SparkMLlib

Answer: c) SparkScheduler

Question: Which Spark API allows you to schedule jobs based on time intervals?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkScheduler



Answer: c) SparkStreaming

Question: Which Spark API is used for working with large-scale unstructured data like text and images?

- a) SparkCore
- b) SparkSQL
- c) SparkNLP
- d) SparkMLlib

Answer: c) SparkNLP

Question: What is the role of the "Cluster Manager" in Spark?

- a) It handles data replication across the cluster.
- b) It optimizes the Spark job execution plan.
- c) It manages resources and schedules Spark applications.
- d) It performs data cleaning operations.

Answer: c) It manages resources and schedules Spark applications.

Question: Which Spark API provides support for distributed matrix operations and linear algebra?

- a) SparkCore
- b) SparkSQL
- c) SparkMatrix
- d) SparkMLlib

Answer: c) SparkMatrix

Question: In Spark, which component is responsible for managing the SparkContext?

- a) SparkCore
- b) SparkContext
- c) SparkDriver
- d) SparkMaster

Answer: b) SparkContext

Question: Which Spark API is used for working with distributed data and parallel processing?

- a) SparkCore
- b) SparkSQL
- c) SparkDistributed
- d) SparkMLlib

Answer: a) SparkCore

Question: What is the primary advantage of using DataFrame over RDD in Spark?

- a) DataFrame offers better fault tolerance.
- b) DataFrame supports distributed machine learning.
- c) DataFrame provides better performance optimization.
- d) DataFrame allows real-time streaming.

Answer: c) DataFrame provides better performance optimization.

Question: Which Spark API is designed for interactive querying and processing semi-structured data?

- a) SparkCore
- b) SparkSQL
- c) SparkStreaming
- d) SparkMLlib

Answer: b) SparkSQL

Question: Which Spark API is used for working with external data sources and data serialization formats?

- a) SparkCore
- b) SparkSQL
- c) SparkFileIO
- d) SparkMLlib

Answer: c) SparkFileIO

Question: What is the primary role of the "BlockManager" in Spark?

- a) It handles data replication across the cluster.
- b) It manages resources and schedules Spark applications.
- c) It performs data cleaning operations.
- d) It manages data storage and caching.

Answer: d) It manages data storage and caching.

Question: Which Spark API provides support for distributed machine learning algorithms and model training?

- a) SparkCore
- b) SparkSQL
- c) MLlib
- d) SparkMLlib

Answer: d) SparkMLlib

Question: Which Spark API is responsible for managing data partitioning and distribution across the cluster?

- a) SparkCore
- b) SparkSQL
- c) SparkDistributed
- d) SparkMLlib

Answer: a) SparkCore

Question: Which Apache Spark library is specifically designed for deep learning tasks?

- a) SparkMLlib
- b) SparkDeep
- c) SparkDL
- d) SparkDLlib

Answer: b) SparkDeep

Question: In Apache Spark, which component is responsible for managing the execution of deep learning tasks?

- a) SparkContext
- b) SparkScheduler
- c) SparkDriver
- d) SparkExecutor

Answer: d) SparkExecutor

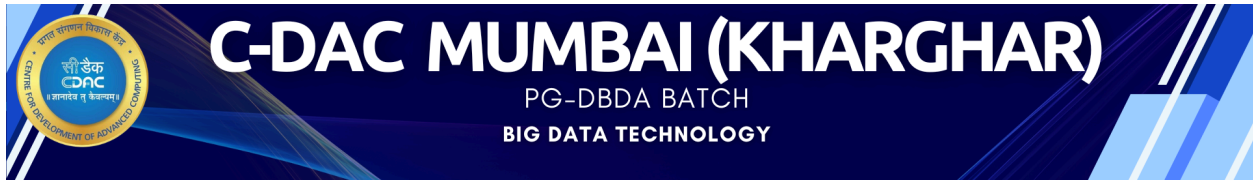
Question: Which programming language is primarily used for implementing deep learning algorithms in Spark?

- a) Java
- b) Scala
- c) Python
- d) R

Answer: c) Python

Question: Which Spark library allows seamless integration with popular deep learning frameworks like TensorFlow and Keras?

- a) SparkDeep
- b) SparkDLlib



- c) DeepSpark
- d) Deeplearning4j

Answer: a) SparkDeep

Question: What is the primary advantage of using Apache Spark for deep learning tasks?

- a) Distributed training across multiple nodes
- b) Faster execution of deep learning models
- c) Built-in support for GPU acceleration
- d) Seamless integration with Hadoop ecosystem

Answer: a) Distributed training across multiple nodes

Question: Which Spark API is commonly used for distributed deep learning tasks?

- a) SparkCore
- b) SparkSQL
- c) SparkDLlib
- d) SparkMLlib

Answer: c) SparkDLlib

Question: Which deep learning architecture is specifically designed for handling sequential data like text and time series?

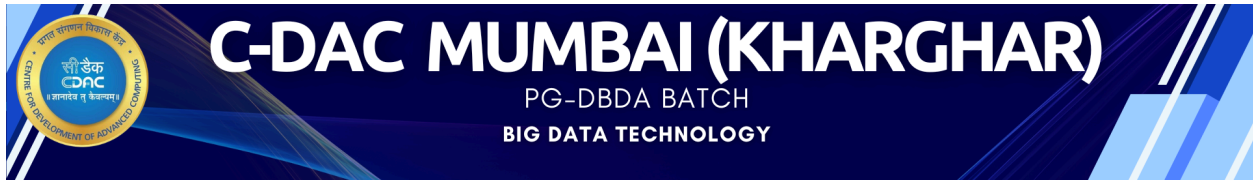
- a) Convolutional Neural Network (CNN)
- b) Recurrent Neural Network (RNN)
- c) Long Short-Term Memory (LSTM)
- d) Generative Adversarial Network (GAN)

Answer: b) Recurrent Neural Network (RNN)

Question: In Spark, which component is responsible for managing data loading and preprocessing for deep learning tasks?

- a) SparkContext
- b) SparkDLlib
- c) SparkDataLoader
- d) SparkTransformer

Answer: c) SparkDataLoader



Question: Which Spark library is developed to leverage distributed GPU resources for deep learning tasks?

- a) SparkDLlib
- b) SparkDeep
- c) DeepSpark
- d) SparkGPU

Answer: c) DeepSpark

Question: Which deep learning algorithm is commonly used for image recognition tasks?

- a) Support Vector Machine (SVM)
- b) k-Nearest Neighbors (k-NN)
- c) Convolutional Neural Network (CNN)
- d) Decision Tree

Answer: c) Convolutional Neural Network (CNN)

Question: In Spark, what is the role of the "ModelCheckpoint" callback in deep learning tasks?

- a) It saves the model checkpoints for resuming training in case of failures.
- b) It optimizes the model's architecture during training.
- c) It performs data shuffling during training.
- d) It applies regularization to the model.

Answer: a) It saves the model checkpoints for resuming training in case of failures.

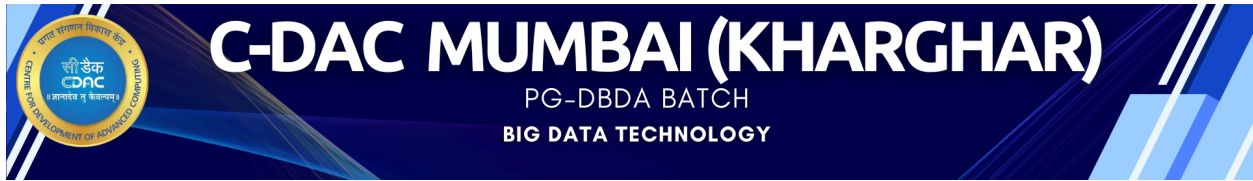
Question: Which Spark library provides support for distributed hyperparameter tuning for deep learning models?

- a) SparkDLlib
- b) SparkTune
- c) HyperSpark
- d) SparkHyperparam

Answer: b) SparkTune

Question: What is the primary benefit of using distributed deep learning with Spark?

- a) Better accuracy of deep learning models
- b) Faster training on large datasets
- c) Lower hardware requirements
- d) Support for cloud-based training



Answer: b) Faster training on large datasets

Question: In Spark, what is the role of the "EarlyStopping" callback in deep learning tasks?

- a) It stops the training process once a certain number of epochs is reached.
- b) It prevents overfitting by stopping training when the model performance plateaus.
- c) It skips the initial epochs for faster convergence.
- d) It adjusts the learning rate dynamically during training.

Answer: b) It prevents overfitting by stopping training when the model performance plateaus.

Question: Which Spark library allows you to perform distributed model evaluation and validation?

- a) SparkDLlib
- b) SparkValidation
- c) SparkMetrics
- d) SparkDeep

Answer: c) SparkMetrics

Question: Which deep learning algorithm is commonly used for natural language processing tasks?

- a) Convolutional Neural Network (CNN)
- b) Recurrent Neural Network (RNN)
- c) Long Short-Term Memory (LSTM)
- d) Support Vector Machine (SVM)

Answer: c) Long Short-Term Memory (LSTM)

Question: In Spark, what is the role of the "LearningRateScheduler" callback in deep learning tasks?

- a) It updates the model's learning rate during training based on a predefined schedule.
- b) It adjusts the batch size dynamically during training.
- c) It handles the data shuffling during training.
- d) It computes the gradients for backpropagation.

Answer: a) It updates the model's learning rate during training based on a predefined schedule.

Question: Which Spark library allows you to build and train deep learning models using a high-level API similar to Keras?

- a) SparkDeep
- b) SparkDLlib
- c) KerasSpark
- d) DeepLearningKeras

Answer: c) KerasSpark

Question: In deep learning, what is the purpose of data augmentation?

- a) To preprocess data for better model convergence
- b) To generate more training data by applying random transformations
- c) To apply dimensionality reduction to the input features
- d) To add noise to the target labels for regularization

Answer: b) To generate more training data by applying random transformations

Question: Which Spark library provides support for distributed model deployment and serving?

- a) SparkServing
- b) SparkDLlib
- c) SparkServingEngine
- d) SparkModelServe

Answer: a) SparkServing

Question: What is the primary advantage of using distributed deep learning for large-scale datasets?

- a) Better generalization of the deep learning model
- b) Faster convergence during training
- c) Reduced risk of overfitting
- d) Improved interpretability of the model

Answer: b) Faster convergence during training

Question: Which deep learning architecture is commonly used for generating new content, such as images or text?

- a) Convolutional Neural Network (CNN)
- b) Recurrent Neural Network (RNN)
- c) Generative Adversarial Network (GAN)
- d) Autoencoder

Answer: c) Generative Adversarial Network (GAN)

Question: In Spark, what is the role of the "Dropout" layer in deep learning models?

- a) It skips a fraction of neurons during training to reduce overfitting.
- b) It initializes the model's weights randomly.
- c) It performs data shuffling during training.
- d) It adjusts the learning rate dynamically during training.

Answer: a) It skips a fraction of neurons during training to reduce overfitting.

Question: Which Spark library provides support for distributed model inference and prediction?

- a) SparkInfer
- b) SparkDLlib
- c) SparkInference
- d) SparkPredict

Answer: c) SparkInference

Question: In deep learning, what is the purpose of transfer learning?

- a) To transfer knowledge from one domain to another
- b) To transfer data from a local storage to a distributed storage system
- c) To transfer gradients during backpropagation
- d) To transfer input features to the output labels

Answer: a) To transfer knowledge from one domain to another

Question: Which deep learning algorithm is commonly used for anomaly detection tasks?

- a) Recurrent Neural Network (RNN)
- b) Convolutional Neural Network (CNN)
- c) Autoencoder
- d) Decision Tree

Answer: c) Autoencoder

Question: In Spark, what is the role of the "Optimizer" in deep learning tasks?

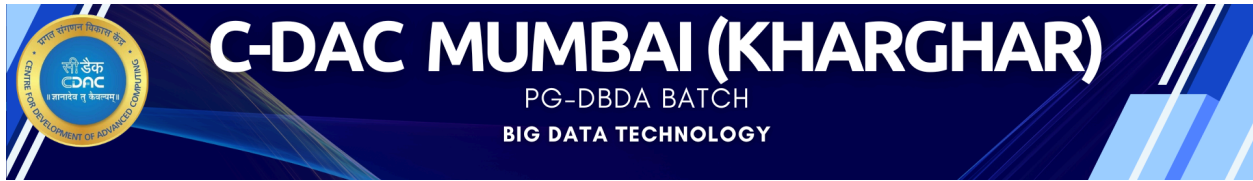
- a) It optimizes the hardware resources for distributed training.
- b) It optimizes the learning rate during training.
- c) It performs data shuffling during training.
- d) It optimizes the model's weights during training.

Answer: d) It optimizes the model's weights during training.

Question: Which Spark library provides support for distributed model monitoring and performance visualization?

- a) SparkDLlib
- b) SparkMonitoring
- c) SparkMetrics
- d) SparkMonitor

Answer: c) SparkMetrics



Question: In deep learning, what is the purpose of batch normalization?

- a) To normalize the target labels for better model convergence
- b) To normalize the input features to have zero mean and unit variance
- c) To normalize the gradients during backpropagation
- d) To normalize the model's weights during training

Answer: b) To normalize the input features to have zero mean and unit variance

Question: Which Spark library provides support for distributed model optimization algorithms like Adam and RMSprop?

- a) SparkOptim
- b) SparkDLlib
- c) SparkOptimizer
- d) SparkOptimization

Answer: c) SparkOptimizer

Question: In Spark, what is the role of the "BatchNormalization" layer in deep learning models?

- a) It normalizes the gradients during backpropagation.
- b) It normalizes the model's weights during training.
- c) It performs data shuffling during training.
- d) It normalizes the input activations to have zero mean and unit variance.

Answer: d) It normalizes the input activations to have zero mean and unit variance.

Question: Which deep learning architecture is commonly used for classification tasks?

- a) Convolutional Neural Network (CNN)
- b) Recurrent Neural Network (RNN)
- c) Long Short-Term Memory (LSTM)
- d) Support Vector Machine (SVM)

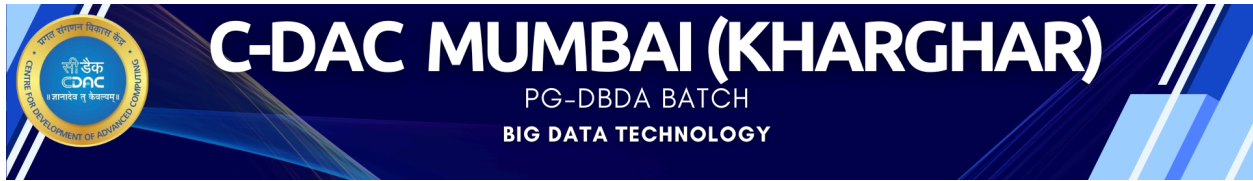
Answer: a) Convolutional Neural Network (CNN)

Question: In Spark, what is the role of the "LearningRateFinder" in deep learning tasks?

- a) It adjusts the learning rate dynamically during training.
- b) It computes the gradients for backpropagation.
- c) It optimizes the model's architecture during training.
- d) It finds the optimal learning rate for the model.

Answer: d) It finds the optimal learning rate for the model.

Question: Which deep learning algorithm is commonly used for recommendation systems?



- a) Convolutional Neural Network (CNN)
- b) Recurrent Neural Network (RNN)
- c) Matrix Factorization
- d) k-Nearest Neighbors (k-NN)

Answer: c) Matrix Factorization

Question: In Spark, what is the role of the "ReLU" activation function in deep learning models?

- a) It performs data shuffling during training.
- b) It optimizes the model's architecture during training.
- c) It adjusts the learning rate dynamically during training.
- d) It introduces non-linearity to the model.

Answer: d) It introduces non-linearity to the model.

Question: Which Spark library provides support for distributed model serialization and deserialization?

- a) SparkDLlib
- b) SparkSerialization
- c) SparkSerialize
- d) SparkModelIO

Answer: a) SparkDLlib

Question: In deep learning, what is the purpose of weight regularization?

- a) To adjust the model's learning rate during training
- b) To regularize the gradients during backpropagation
- c) To apply random weights initialization
- d) To prevent overfitting by penalizing large weights

Answer: d) To prevent overfitting by penalizing large weights

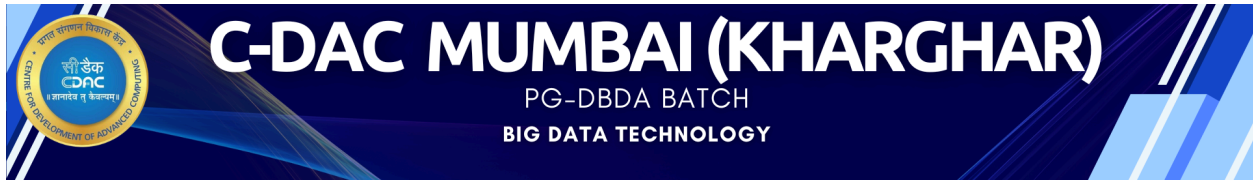
Question: Which deep learning algorithm is commonly used for sentiment analysis tasks?

- a) Convolutional Neural Network (CNN)
- b) Recurrent Neural Network (RNN)
- c) Long Short-Term Memory (LSTM)
- d) Naive Bayes

Answer: b) Recurrent Neural Network (RNN)

Question: In Spark, what is the role of the "DataParallelism" technique in deep learning tasks?

- a) It optimizes the model's architecture during training.
- b) It performs data shuffling during training.



- c) It processes data in parallel across multiple nodes.
- d) It adjusts the learning rate dynamically during training.

Answer: c) It processes data in parallel across multiple nodes.

Question: Which Spark library provides support for distributed model tuning and optimization?

- a) SparkDLlib
- b) SparkOptim
- c) SparkTune
- d) SparkHyperparam

Answer: c) SparkTune

Question: In deep learning, what is the purpose of the "Dropout" regularization technique?

- a) To apply data augmentation during training
- b) To drop a fraction of neurons during training to prevent overfitting
- c) To drop the last few layers of the model
- d) To drop the output features during training

Answer: b) To drop a fraction of neurons during training to prevent overfitting

Question: Which deep learning architecture is commonly used for speech recognition tasks?

- a) Convolutional Neural Network (CNN)
- b) Recurrent Neural Network (RNN)
- c) Long Short-Term Memory (LSTM)
- d) Transformer

Answer: b) Recurrent Neural Network (RNN)

Question: In Spark, what is the role of the "GradientClipping" technique in deep learning tasks?

- a) It optimizes the model's architecture during training.
- b) It performs data shuffling during training.
- c) It adjusts the learning rate dynamically during training.
- d) It clips the gradients to prevent exploding gradients during backpropagation.

Answer: d) It clips the gradients to prevent exploding gradients during backpropagation.

Question: Which Spark library provides support for distributed model ensemble techniques?

- a) SparkDLlib
- b) SparkEnsemble
- c) SparkEnsembleML
- d) SparkEnsembleLearning

Answer: c) SparkEnsembleML

Question: In deep learning, what is the purpose of the "BatchNormalization" technique?

- a) To adjust the batch size dynamically during training
- b) To normalize the input activations to have zero mean and unit variance
- c) To normalize the gradients during backpropagation
- d) To normalize the model's weights during training

Answer: b) To normalize the input activations to have zero mean and unit variance

Question: Which deep learning algorithm is commonly used for time series forecasting?

- a) Convolutional Neural Network (CNN)
- b) Recurrent Neural Network (RNN)
- c) Long Short-Term Memory (LSTM)
- d) Linear Regression

Answer: c) Long Short-Term Memory (LSTM)

Question: In Spark, what is the role of the "LearningRateDecay" callback in deep learning tasks?

- a) It adjusts the learning rate dynamically during training.
- b) It optimizes the model's architecture during training.
- c) It performs data shuffling during training.
- d) It decays the learning rate over time to improve convergence.

Answer: d) It decays the learning rate over time to improve convergence.

Question: Which deep learning architecture is commonly used for natural language translation tasks?

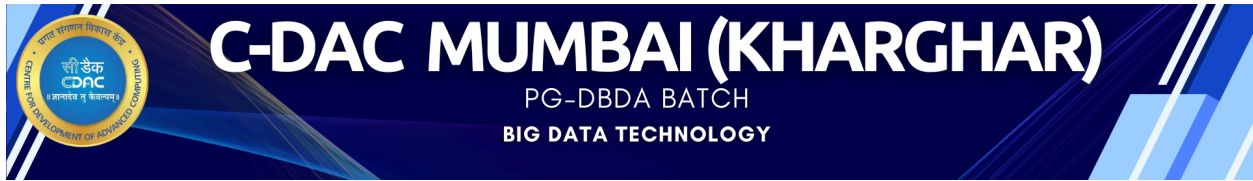
- a) Convolutional Neural Network (CNN)
- b) Recurrent Neural Network (RNN)
- c) Transformer
- d) Autoencoder

Answer: c) Transformer

Question: In Spark, what is the role of the "EarlyStopping" technique in deep learning tasks?

- a) It stops the training process once a certain number of epochs is reached.
- b) It prevents overfitting by stopping training when the model performance plateaus.
- c) It skips the initial epochs for faster convergence.
- d) It adjusts the learning rate dynamically during training.

Answer: b) It prevents overfitting by stopping training when the model performance plateaus.



Question: Which Spark library provides support for distributed model explainability and interpretability?

- a) SparkDLlib
- b) SparkInterpret
- c) SparkExplainer
- d) SparkExplainable

Answer: b) SparkInterpret

Question:

What is the primary function of the Spark Driver in a Spark Streaming application?

- A) Manage Spark Executors
- B) Process the streaming data
- C) Schedule streaming tasks
- D) Execute Spark transformations

Answer: C) Schedule streaming tasks

Explanation: The Spark Driver is responsible for dividing the streaming data into batches and scheduling tasks to Spark Executors for processing.

Question:

Which streaming source does Spark Streaming NOT support out of the box?

- A) Kafka
- B) Flume
- C) Amazon S3
- D) RabbitMQ

Answer: D) RabbitMQ

Explanation: Spark Streaming natively supports integration with Kafka, Flume, and Amazon S3 as data sources, but RabbitMQ integration requires additional customizations.

Question:

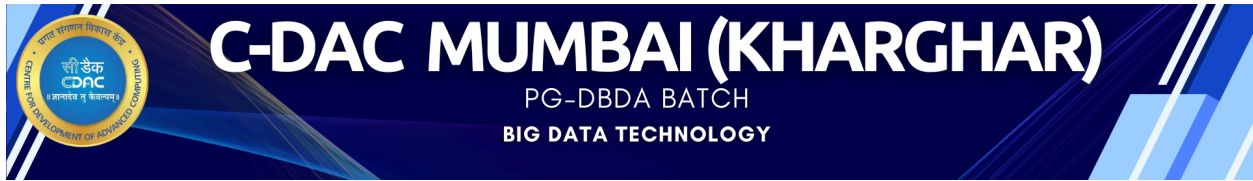
In Spark Streaming, what does the term "micro-batch" refer to?

- A) A small batch of data processed every 10 seconds
- B) The minimum time interval between consecutive batches
- C) A batch of data processed in parallel across all executors
- D) A batch of data with a size of less than 1 MB

Answer: B) The minimum time interval between consecutive batches

Explanation: Spark Streaming processes data in micro-batches, which are small chunks of data processed at regular intervals.

Question:



Which Spark Streaming transformation is used to aggregate data over a sliding window?

- A) reduceByKey
- B) updateStateByKey
- C) window
- D) aggregateByKey

Answer: C) window

Explanation: The window transformation is used for aggregating data over a sliding window of time in Spark Streaming.

Question:

What is the significance of using Spark's updateStateByKey transformation in a streaming application?

- A) It enables stateful processing, allowing data to be accumulated over time.
- B) It improves the fault-tolerance of the application by checkpointing data regularly.
- C) It enhances the scalability of the application by distributing data across multiple nodes.
- D) It simplifies data processing by automatically handling schema evolution.

Answer: A) It enables stateful processing, allowing data to be accumulated over time.

Explanation: updateStateByKey allows you to maintain stateful information across batches in Spark Streaming, enabling the accumulation of data over time.

Question:

Which Spark Streaming window operation combines data from multiple windows into a single RDD?

- A) reduceByKeyAndWindow
- B) countByWindow
- C) updateStateByKey
- D) reduceByKey

Answer: A) reduceByKeyAndWindow

Explanation: reduceByKeyAndWindow performs reduction over a sliding window, combining data from multiple windows into a single RDD.

Question:

When does data in a Spark Streaming application get processed in parallel?

- A) During the data ingestion phase
- B) During the data transformation phase
- C) During the data output phase
- D) Only when checkpointing is enabled

Answer: B) During the data transformation phase

Explanation: Spark Streaming processes data in parallel during the data transformation phase, which involves applying various transformations on the data.

Question:

What is the purpose of watermarking in Spark Streaming?

- A) To eliminate duplicate data from the stream
- B) To efficiently handle late-arriving data in event time processing
- C) To balance the load across multiple executors
- D) To synchronize data processing across multiple micro-batches

Answer: B) To efficiently handle late-arriving data in event time processing

Explanation: Watermarking is used to manage event time data in Spark Streaming and efficiently handle late-arriving data within a specified window.

Question:

Which Spark Streaming output mode allows a streaming application to write data to external storage in batches?

- A) Complete Mode
- B) Append Mode
- C) Update Mode
- D) Continuous Mode

Answer: B) Append Mode

Explanation: Append Mode allows a streaming application to write data to external storage in batches, where each batch is a complete micro-batch.

Question:

In a Spark Streaming application, what happens if the processing time of a micro-batch exceeds the batch interval?

- A) The application crashes with an OutOfMemoryError.
- B) The excess data is dropped, leading to data loss.
- C) The batch interval is automatically increased to accommodate processing time.
- D) The next micro-batch is processed immediately without waiting for the previous one to complete.

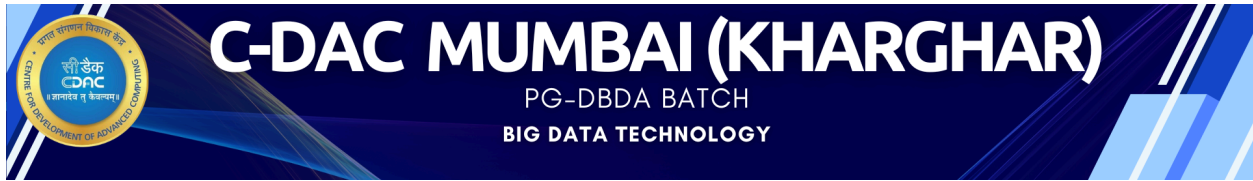
Answer: D) The next micro-batch is processed immediately without waiting for the previous one to complete.

Explanation: Spark Streaming automatically moves on to the next micro-batch if the processing time exceeds the batch interval, which may result in overlapping processing.

Question:

Which Spark Streaming input source provides built-in support for backpressure?

- A) Kafka
- B) Flume
- C) Amazon Kinesis



D) Twitter Streaming

Answer: C) Amazon Kinesis

Explanation: Amazon Kinesis provides built-in support for backpressure, which helps to control data flow between the producer and the consumer in Spark Streaming.

Question:

What is the recommended approach to achieve exactly-once processing in Spark Streaming?

- A) Enable Spark Streaming's at-least-once processing and use idempotent operations in transformations.
- B) Use exactly-once semantics offered by the streaming source, such as Kafka.
- C) Use Spark Streaming's checkpointing feature and disable at-least-once processing.
- D) Exactly-once processing is not possible in Spark Streaming.

Answer: A) Enable Spark Streaming's at-least-once processing and use idempotent operations in transformations.

Explanation: Exactly-once processing is not guaranteed in Spark Streaming, but enabling at-least-once processing and using idempotent operations can achieve close to exactly-once semantics.

Question:

Which Spark Streaming window operation supports processing time-based windows?

- A) reduceByKeyAndWindow
- B) countByWindow
- C) window
- D) reduceByKey

Answer: C) window

Explanation: The window operation supports both event time-based and processing time-based windows in Spark Streaming.

Question:

What is the purpose of the setLogLevel method in a Spark Streaming application?

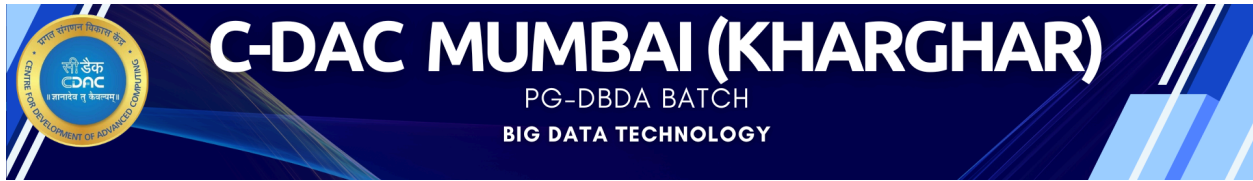
- A) To specify the level of parallelism for data processing
- B) To set the level of verbosity for log messages
- C) To enable/disable checkpointing for fault tolerance
- D) To configure the streaming batch interval

Answer: B) To set the level of verbosity for log messages

Explanation: The setLogLevel method is used to set the level of verbosity for log messages in Spark Streaming.

Question:

In Spark Streaming, which method is used to create a stream from custom data sources?



- A) streamFromSource
- B) createStream
- C) createDStream
- D) fromCustomSource

Answer: C) createDStream

Explanation: The createDStream method is used to create a DStream (Discretized Stream) from custom data sources in Spark Streaming.

Question:

Which Spark Streaming output mode allows a streaming application to continuously write data to external storage?

- A) Complete Mode
- B) Append Mode
- C) Update Mode
- D) Continuous Mode

Answer: D) Continuous Mode

Explanation: Continuous Mode enables a streaming application to continuously write data to external storage, providing low-latency, near-real-time processing.

Question:

What happens if a Spark Streaming application crashes and checkpointing is not enabled?

- A) The application resumes processing from the last micro-batch automatically.
- B) The application starts processing from the beginning of the stream.
- C) The application crashes with an error, and data is lost.
- D) The application prompts the user to provide a new starting offset.

Answer: B) The application starts processing from the beginning of the stream.

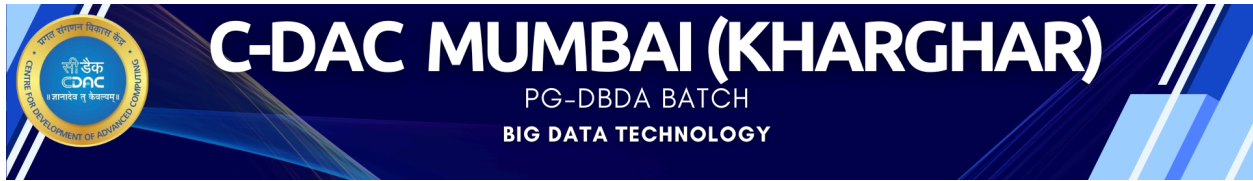
Explanation: Without checkpointing, Spark Streaming cannot recover the processing state after a crash, and the application starts processing from the beginning of the stream.

Question:

Which Spark Streaming transformation is used to group data by a key and apply a function to the grouped data?

- A) map
- B) groupBy
- C) reduceByKey
- D) transform

Answer: C) reduceByKey



Explanation: The reduceByKey transformation groups data by key and applies a reduction function in Spark Streaming.

Question:

What is the minimum number of Spark Executors required to run a Spark Streaming application?

- A) 1
- B) 2
- C) 3
- D) It depends on the size of the input data.

Answer: A) 1

Explanation: A single Spark Executor is the minimum required to run a Spark Streaming application, although more Executors can be utilized for parallel processing.

Question:

Which data serialization format is preferred in Spark Streaming for better performance?

- A) JSON
- B) XML
- C) Avro
- D) CSV

Answer: C) Avro

Explanation: Avro is a binary data serialization format that is preferred in Spark Streaming for better performance due to its efficient storage and processing characteristics.

Question:

In Spark Streaming, what is the role of the "driver" and "executor" nodes?

- A) Both driver and executor nodes process the streaming data.
- B) The driver node is responsible for job scheduling, and executor nodes process the data.
- C) The driver node stores the stream state, while executor nodes handle data processing.
- D) The driver node acts as a sink, and executor nodes are the data sources.

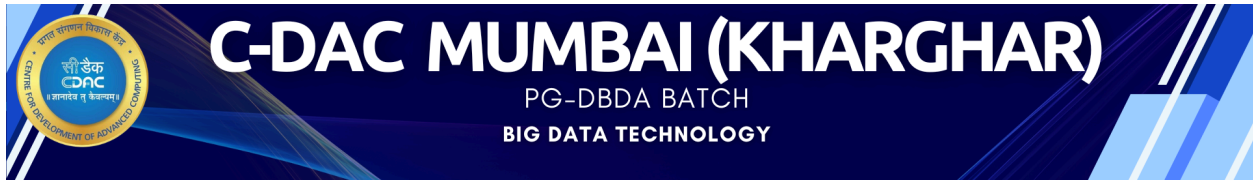
Answer: B) The driver node is responsible for job scheduling, and executor nodes process the data.

Explanation: The driver node is responsible for job scheduling, managing the stream metadata, and dividing the streaming data into micro-batches. Executor nodes are responsible for processing the data in parallel.

Question:

Which window operation in Spark Streaming allows overlapping windows?

- A) window
- B) reduceByKeyAndWindow
- C) countByWindow



D) updateStateByKey

Answer: B) reduceByKeyAndWindow

Explanation: The reduceByKeyAndWindow operation allows the use of overlapping windows in Spark Streaming, which means some data points can belong to multiple windows.

Question:

What is the purpose of enabling "backpressure" in Spark Streaming?

- A) To reduce the load on the Spark cluster during peak times
- B) To avoid data loss during data ingestion from Kafka
- C) To throttle the rate of data input to match the processing rate
- D) To allow Spark to prioritize certain micro-batches over others

Answer: C) To throttle the rate of data input to match the processing rate

Explanation: Enabling backpressure helps control the rate of data ingestion to match the processing rate in Spark Streaming, avoiding backlogs and resource exhaustion.

Question:

Which Spark Streaming transformation allows you to join data from two DStreams?

- A) join
- B) transform
- C) updateStateByKey
- D) union

Answer: A) join

Explanation: The join transformation in Spark Streaming allows you to join data from two DStreams based on a common key.

Question:

What is the advantage of using the updateStateByKey transformation over other transformations in Spark Streaming?

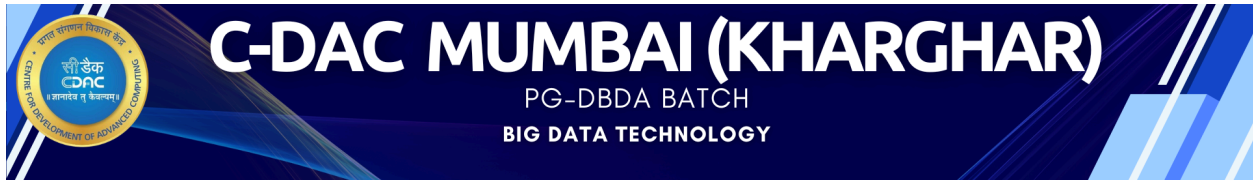
- A) It provides better performance due to in-memory processing.
- B) It enables fault-tolerance through data replication.
- C) It allows stateful processing and accumulates data over time.
- D) It supports data partitioning for efficient processing.

Answer: C) It allows stateful processing and accumulates data over time.

Explanation: The updateStateByKey transformation enables stateful processing in Spark Streaming, allowing data to be accumulated over time across micro-batches.

Question:

How does Spark Streaming handle data processing failures within a micro-batch?



- A) Spark Streaming automatically retries the failed tasks until successful.
 - B) The entire micro-batch is discarded, and the application continues processing the next batch.
 - C) Only the failed tasks are re-executed within the micro-batch.
 - D) Spark Streaming stops processing the data and waits for manual intervention.
- Answer: B) The entire micro-batch is discarded, and the application continues processing the next batch.

Explanation: In case of data processing failures within a micro-batch, Spark Streaming discards the entire micro-batch and continues processing the next batch.

Question:

In Spark Streaming, what happens if the input data size is smaller than the batch interval?

- A) Spark Streaming buffers the data until it reaches the batch interval size.
- B) Spark Streaming processes the available data immediately, even if it's smaller.
- C) The application throws an error and stops processing.
- D) The batch interval is automatically reduced to match the data size.

Answer: B) Spark Streaming processes the available data immediately, even if it's smaller.

Explanation: If the input data size is smaller than the batch interval, Spark Streaming processes the available data immediately without waiting for additional data.

Question:

What is the purpose of using checkpointing in Spark Streaming?

- A) To improve the application's performance by caching data.
 - B) To maintain the application state and ensure fault tolerance.
 - C) To save the final results of the stream processing.
 - D) To partition the data efficiently across multiple executors.
- Answer: B) To maintain the application state and ensure fault tolerance.

Explanation: Checkpointing in Spark Streaming is used to maintain the application state, ensuring fault tolerance in case of failures.

Question:

Which Spark Streaming output mode allows the streaming application to output only the changed records in the result?

- A) Complete Mode
- B) Append Mode
- C) Update Mode
- D) Continuous Mode

Answer: C) Update Mode

Explanation: In Update Mode, the streaming application outputs only the changed records in the result, indicating the update to the previous state.

Question:

What is the default checkpoint directory in Spark Streaming?

- A) /spark/checkpoints
- B) /tmp/spark/checkpoints
- C) /user/spark/checkpoints
- D) /var/spark/checkpoints

Answer: B) /tmp/spark/checkpoints

Explanation: The default checkpoint directory in Spark Streaming is /tmp/spark/checkpoints.

Question:

Which Spark Streaming window operation is suitable for counting the number of elements in a sliding window?

- A) reduceByKeyAndWindow
- B) countByWindow
- C) window
- D) reduceByKey

Answer: B) countByWindow

Explanation: The countByWindow operation is suitable for counting the number of elements in a sliding window in Spark Streaming.

Question:

What is the purpose of using the foreachRDD transformation in Spark Streaming?

- A) To apply a function to each element of the DStream.
- B) To filter the data before processing.
- C) To repartition the data for better parallelism.
- D) To convert the DStream to a DataFrame.

Answer: A) To apply a function to each element of the DStream.

Explanation: The foreachRDD transformation allows you

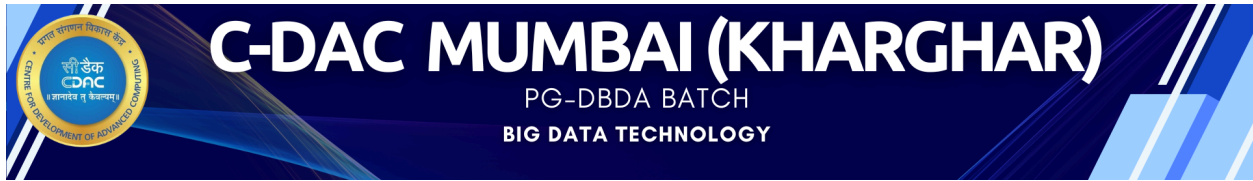
Question:

What is the role of the Metastore in Apache Hive?

- a) It stores the Hive data warehouse files on HDFS.
- b) It stores metadata such as table schemas, partitions, and table locations.
- c) It manages the MapReduce jobs for Hive queries.
- d) It acts as a distributed file system for storing query results.

Answer: b) It stores metadata such as table schemas, partitions, and table locations.

Question:



Which of the following statements about partitioning in Hive is correct?

- a) Partitioning improves data integrity but has no impact on query performance.
- b) Partitioning can only be done on numerical columns.
- c) Partitioning allows data to be stored in subdirectories based on the partitioning column.
- d) Partitioning can only be done on tables with a single column.

Answer: c) Partitioning allows data to be stored in subdirectories based on the partitioning column.

Question:

In Hive, what does the term "bucketing" refer to?

- a) Dividing data into smaller parts to improve query performance.
- b) Randomly shuffling data across nodes in a Hadoop cluster.
- c) Storing data in separate tables to enhance security.
- d) Grouping data based on a column into a fixed number of buckets to improve join performance.

Answer: d) Grouping data based on a column into a fixed number of buckets to improve join performance.

Question:

Which file format is the most suitable for large-scale data processing in Hive due to its support for compression and splittable input?

- a) JSON
- b) XML
- c) Parquet
- d) Avro

Answer: c) Parquet

Question:

What happens when you run a Hive query with "SET hive.mapred.mode=strict;"?

- a) It restricts the query to run only in strict mode where it will use MapReduce exclusively.
- b) It enables the strict type-checking for Hive queries to avoid type-related errors.
- c) It forces the Hive query to execute in local mode on the Hive server.
- d) It sets a strict limit on the number of MapReduce jobs a query can launch.

Answer: a) It restricts the query to run only in strict mode where it will use MapReduce exclusively.

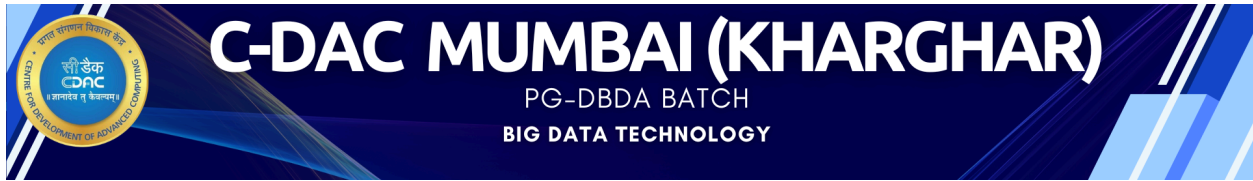
Question:

In Hive, what is the purpose of the EXPLAIN keyword before a SELECT query?

- a) It displays the execution plan of the query without actually running it.
- b) It instructs the Hive optimizer to use a different execution plan.
- c) It enables debug mode for the SELECT query.
- d) It forces the query to execute on a specific node in the cluster.

Answer: a) It displays the execution plan of the query without actually running it.

Question:



Which component in Hive is responsible for translating HiveQL queries into a series of MapReduce jobs?

- a) Hive Client
- b) Hive Metastore
- c) Hive Driver
- d) Hive Compiler

Answer: c) Hive Driver

Question:

What is the primary benefit of using Hive over traditional SQL databases for analytical processing?

- a) Hive supports real-time data processing, while SQL databases do not.
- b) Hive is more cost-effective and does not require any hardware infrastructure.
- c) Hive can handle large-scale data processing in a distributed computing environment.
- d) Hive has built-in machine learning algorithms for predictive analysis.

Answer: c) Hive can handle large-scale data processing in a distributed computing environment.

What is Apache Spark's primary programming abstraction for handling large-scale data processing?

- a) MapReduce
- b) DataFrames
- c) SparkSQL
- d) RDDs

Answer: d) RDDs

Which Spark component enables real-time stream processing?

- a) Spark MLlib
- b) Spark Streaming
- c) Spark GraphX
- d) Spark Structured Streaming

Answer: d) Spark Structured Streaming

What is the default micro-batch interval in Spark Structured Streaming?

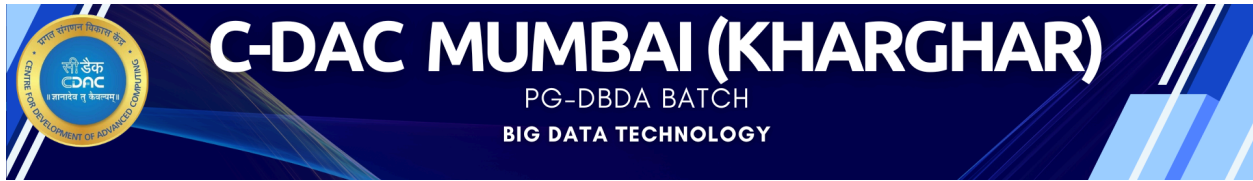
- a) 1 second
- b) 2 seconds
- c) 5 seconds
- d) 10 seconds

Answer: c) 5 seconds

Which Spark function allows you to apply a transformation on each element of an RDD?

- a) reduce
- b) map
- c) filter
- d) collect

Answer: b) map



In Spark Streaming, what does "windowing" refer to?

- a) Dividing the input data into fixed-size batches
- b) Sliding the window over the data stream to perform computations on a set of data points
- c) Aggregating data into a single output at the end of each micro-batch
- d) Dynamically resizing the Spark cluster based on the data load

Answer: b) Sliding the window over the data stream to perform computations on a set of data points

What is the primary advantage of using Spark Structured Streaming over Spark Streaming?

- a) Lower latency
 - b) Higher throughput
 - c) Simplified API with DataFrame operations
 - d) Support for micro-batch processing
- Answer: c) Simplified API with DataFrame operations

Which of the following is NOT a supported data source for Spark Structured Streaming?

- a) Kafka
- b) HDFS
- c) Amazon S3
- d) MySQL

Answer: d) MySQL

What is "watermarking" in Spark Structured Streaming?

- a) Adding timestamps to the data stream for synchronization
- b) Cleaning up outdated stateful data in the stream
- c) Resolving conflicts in micro-batch processing
- d) Adjusting the parallelism of the Spark cluster

Answer: b) Cleaning up outdated stateful data in the stream

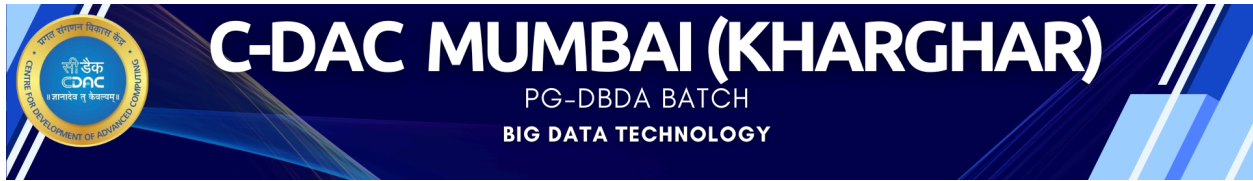
Which Spark operation is used to count the number of elements in an RDD?

- a) countByKey
- b) count
- c) countByValue
- d) length

Answer: b) count

What is the primary benefit of using a stateful transformation in Spark Structured Streaming?

- a) Reduced memory usage
 - b) Improved fault tolerance
 - c) Better performance
 - d) Ability to maintain state across micro-batches
- Answer: d) Ability to maintain state across micro-batches



Which Spark transformation allows you to combine the elements of an RDD into a single value?

- a) fold
- b) reduceByKey
- c) collect
- d) aggregate

Answer: a) fold

In Spark, what is the purpose of the "checkpoint" directory?

- a) Storing intermediate data during a Spark job
- b) Enabling fault-tolerance in Spark Streaming
- c) Caching the data for faster processing
- d) Identifying the Spark cluster leader node

Answer: b) Enabling fault-tolerance in Spark Streaming

Which Spark deployment mode allows multiple applications to share the same cluster?

- a) Client mode
- b) Cluster mode
- c) Local mode
- d) Standalone mode

Answer: d) Standalone mode

Which of the following is true about "Spark Executors"?

- a) Executors are the driver program in Spark.
- b) Executors are responsible for scheduling tasks on the cluster.
- c) Executors are responsible for executing tasks in parallel.
- d) Executors store the metadata of the Spark application.

Answer: c) Executors are responsible for executing tasks in parallel.

What happens if a Spark Executor fails while processing data?

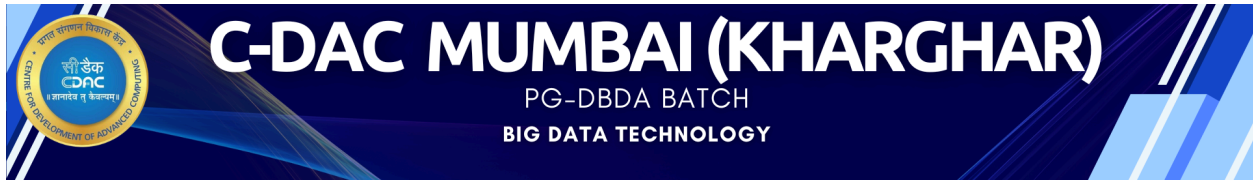
- a) The entire Spark application fails.
- b) The data processed by the failed Executor is lost.
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Answer: b) The data processed by the failed Executor is lost.

What is the role of a "Spark Driver" in a Spark application?

- a) Distributing data across the cluster.
- b) Coordinating tasks and executing Spark jobs.
- c) Executing tasks in parallel on the cluster.
- d) Writing the final results to external storage.

Answer: b) Coordinating tasks and executing Spark jobs.



Which Spark transformation allows you to filter elements of an RDD based on a condition?

- a) map
- b) reduce
- c) filter
- d) collect

Answer: c) filter

What is the default serialization format used by Spark for transferring data between nodes in a cluster?

- a) XML
- b) JSON
- c) Avro
- d) Java Object Serialization (Java Serialization)

Answer: d) Java Object Serialization (Java Serialization)

What is the primary difference between "groupByKey" and "reduceByKey" transformations in Spark?

- a) "groupByKey" shuffles the data, but "reduceByKey" does not.
- b) "reduceByKey" shuffles the data, but "groupByKey" does not.
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Answer: c) "groupByKey" performs a global reduction, while "reduceByKey" performs a per-key reduction.

Which Spark operation is used to remove duplicate elements from an RDD?

- a) distinct
- b) unique
- c) removeDuplicates
- d) deduplicate

Answer: a) distinct

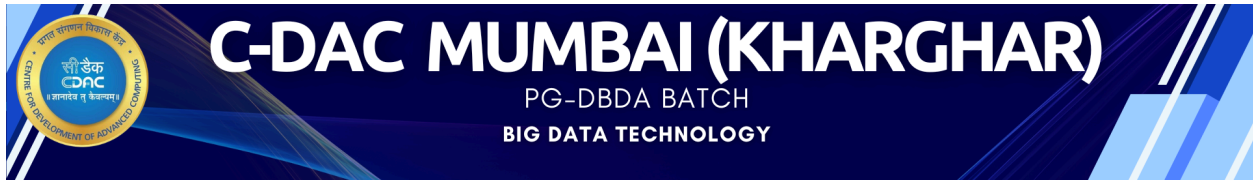
What is the primary advantage of using the Spark standalone cluster manager over other cluster managers like YARN or Mesos?

- a) Better integration with cloud platforms
- b) Simplicity and ease of setup
- c) Support for dynamic scaling of cluster resources
- d) Enhanced security features

Answer: b) Simplicity and ease of setup

In Spark Streaming, what is the significance of the "receiver" component?

- a) It receives and processes data from Kafka topics.
- b) It receives data from the input sources and stores it in memory for processing.
- c) It receives processed data and writes it to external storage.



d) It receives and processes data from Spark executors.

Answer: b) It receives data from the input sources and stores it in memory for processing.

Which Spark operation allows you to combine two RDDs with the same number of partitions into a single RDD?

- a) join
- b) union
- c) merge
- d) zip

Answer: d) zip

What does the "shuffle" operation do in Spark?

- a) Sorts the data based on a specific key.
- b) Groups the data based on a specific key.
- c) Repartitions the data across the cluster.
- d) Transfers data between Spark executors.

Answer: c) Repartitions the data across the cluster.

What is the purpose of Spark's "accumulator" variable?

- a) Storing data in a distributed cache for faster access.
- b) Summing up values across Spark tasks in a fault-tolerant manner.
- c) Controlling the parallelism level of a Spark application.
- d) Tracking the progress of a Spark job.

Answer: b) Summing up values across Spark tasks in a fault-tolerant manner.

What is the function of a "checkpoint" in Spark Structured Streaming?

- a) It acts as a temporary buffer for data to handle network latency.
- b) It stores intermediate data to optimize iterative algorithms.
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How can you achieve exactly-once processing semantics in Spark Structured Streaming?

- a) By using Kafka as the only data source.
- b) By configuring the Spark Streaming engine for at-least-once processing.
- c) By using watermarking and checkpointing.
- d) By avoiding any stateful transformations in the application.

Answer: c) By using watermarking and checkpointing.

What is the primary purpose of Spark's "broadcast variables"?

- a) To share read-only variables across Spark tasks efficiently.
- b) To transfer large files between Spark executors.



- c) To enable communication between the driver program and the executors.
- d) To synchronize the execution of Spark tasks.

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How does Spark Structured Streaming handle late data arrival in a streaming application?

- a) It discards the late data.
- b) It stores the late data in a separate archive for future processing.
- c) It applies watermarking to deal with late data.
- d) It rolls back the entire streaming application to a previous state.

Answer: c) It applies watermarking to deal with late data.

In Spark, what is the primary difference between "coarse-grained" and "fine-grained" operations?

- a) Coarse-grained operations involve data shuffling, while fine-grained operations do not.
- b) Fine-grained operations are faster than coarse-grained operations.
- c) Coarse-grained operations are only available in Spark Streaming, while fine-grained operations are available in batch processing.
- d) Fine-grained operations have a higher level of parallelism than coarse-grained operations.

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What is the role of the "SparkContext" object in a Spark application?

- a) Managing the scheduling of Spark jobs.
- b) Coordinating communication between Spark executors.
- c) Providing access to Spark functionality and resources.
- d) Storing the intermediate data during Spark job execution.

Answer: c) Providing access to Spark functionality and resources.

Which of the following is NOT a supported data sink in Spark Structured Streaming?

- a) HDFS
- b) Kafka
- c) Amazon S3
- d) MongoDB

Answer: d) MongoDB

What is the purpose of Spark's "window" function in Spark SQL?

- a) To perform aggregation over a sliding window of data.
- b) To filter data based on a specific condition.
- c) To join two DataFrames based on a common key.
- d) To create a new DataFrame by combining multiple DataFrames.

Answer: a) To perform aggregation over a sliding window of data.

How does Spark handle data partitioning and distribution across nodes in a cluster?

- a) It uses dynamic partitioning based on the input data size.

- b) It automatically distributes data evenly across all nodes in the cluster.
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What is the main purpose of Spark's "DataFrame" abstraction?

- a) To allow developers to work with structured data in a tabular format.
- b) To provide support for iterative machine learning algorithms.
- c) To enable batch processing of large datasets.
- d) To allow developers to interact with Spark through SQL queries.

Answer: a) To allow developers to work with structured data in a tabular format.

What is the significance of the "driver memory" configuration in a Spark application?

- a) It controls the amount of memory allocated to Spark Executors.
- b) It specifies the amount of memory allocated to the Spark Driver program.
- c) It controls the amount of memory allocated to the cluster manager.
- d) It determines the total memory available for the Spark application.

Answer: b) It specifies the amount of memory allocated to the Spark Driver program.

How can you optimize the performance of a Spark Streaming application?

- a) By increasing the micro-batch interval to minimize overhead.
- b) By avoiding the use of stateful transformations.
- c) By configuring a large number of Spark Executors to increase parallelism.
- d) By reducing the number of partitions in the input data.

Answer: c) By configuring a large number of Spark Executors to increase parallelism.

What does the "checkpoint directory" in Spark Streaming allow you to do?

- a) To store the final results of a Spark job.
- b) To store intermediate data during iterative processing.
- c) To enable fault-tolerance by persisting the metadata of the streaming application.
- d) To schedule periodic checkpoints to improve performance.

Answer: c) To enable fault-tolerance by persisting the metadata of the streaming application.

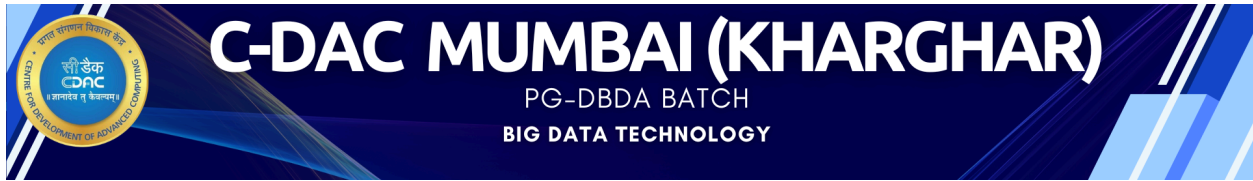
Which of the following is NOT a Spark Streaming source that can be used to ingest data?

- a) Kafka
- b) Amazon S3
- c) Flume
- d) HDFS

Answer: d) HDFS

In Spark, what is the function of the "cache" and "persist" operations?

- a) To store data in memory for faster access.



- b) To compress data for efficient storage.
- c) To shuffle data across the cluster for parallel processing.
- d) To checkpoint data for fault tolerance.

Answer: a) To store data in memory for faster access.

Which Spark operation is used to compute the sum of values in an RDD?

- a) sum
- b) reduce
- c) aggregate
- d) fold

Answer: b) reduce

What is the primary benefit of using Spark Streaming instead of traditional batch processing?

- a) Spark Streaming allows you to process data in real-time.
- b) Spark Streaming offers better fault-tolerance capabilities.
- c) Spark Streaming guarantees exactly-once processing semantics.
- d) Spark Streaming supports a wider variety of data sources.

Answer: a) Spark Streaming allows you to process data in real-time.

What is the function of the "transform" operation in Spark Structured Streaming?

- a) To map the elements of an RDD to a new RDD.
- b) To perform data aggregation over a window of time.
- c) To apply custom, non-aggregate logic on the DataFrames.
- d) To join two DataFrames based on a common key.

Answer: c) To apply custom, non-aggregate logic on the DataFrames.

How does Spark Structured Streaming handle schema evolution when processing data streams?

- a) It automatically adapts to schema changes without any manual intervention.
- b) It requires you to manually update the schema before processing new data.
- c) It automatically applies the same schema used during the initial processing to all subsequent data.
- d) It rejects any data with schema changes to maintain consistency.

Answer: a) It automatically adapts to schema changes without any manual intervention.

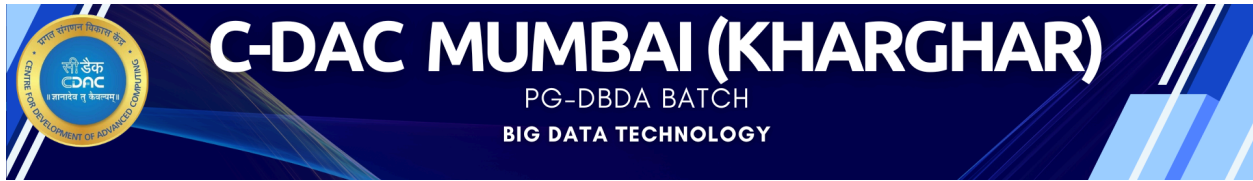
In Spark Structured Streaming, what is the role of the "output mode"?

- a) To control how the final output is written to external storage.
- b) To specify the data format of the input sources.
- c) To define the windowing interval for processing data.
- d) To determine the order of data processing in the streaming application.

Answer: a) To control how the final output is written to external storage.

How does Spark handle data serialization and deserialization?

- a) Spark uses Java Object Serialization by default.



- b) Spark automatically selects the most efficient serialization format based on the data type.
- c) Users can configure the serialization format manually.
- d) Spark always uses Avro for serialization and deserialization.

Answer: a) Spark uses Java Object Serialization by default.

Which Spark operation is used to find the maximum value in an RDD?

- a) max
- b) maximum
- c) reduce
- d) collect

Answer: a) max

What is the role of the "executor memory" configuration in Spark?

- a) It controls the amount of memory allocated to Spark Executors.
- b) It specifies the amount of memory allocated to the Spark Driver program.
- c) It controls the amount of memory allocated to the cluster manager.
- d) It determines the total memory available for the Spark application.

Answer: a) It controls the amount of memory allocated to Spark Executors.

What does "RDD lineage" refer to in Spark?

- a) The transformation of an RDD into a DataFrame.
- b) The sequence of transformations applied to an RDD to calculate its data.
- c) The mapping of an RDD to the physical memory of Spark Executors.
- d) The replication of an RDD across multiple nodes in the cluster.

Answer: b) The sequence of transformations applied to an RDD to calculate its data.

Which Spark operation is used to perform element-wise addition of two RDDs?

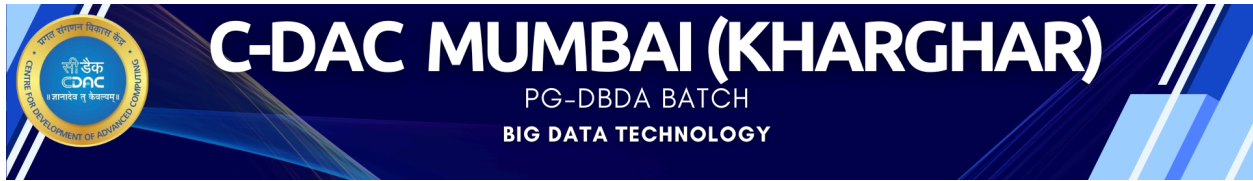
- a) add
- b) plus
- c) union
- d) zip

Answer: d) zip

Question:

Which Hive QL statement should be used to create an external table that points to an existing HDFS location?

- a) `CREATE EXTERNAL TABLE table_name (column1 INT, column2 STRING) LOCATION '/hdfs/path/';`
- b) `CREATE TABLE table_name (column1 INT, column2 STRING) LOCATION '/hdfs/path/';`
- c) `CREATE EXTERNAL TABLE table_name AS (SELECT * FROM data_table);`
- d) `CREATE TABLE table_name AS EXTERNAL '/hdfs/path/' (column1 INT, column2 STRING);`



Answer: a) CREATE EXTERNAL TABLE table_name (column1 INT, column2 STRING) LOCATION '/hdfs/path/';

Question:

In Hive QL, what will happen if you try to insert data into a partition that does not exist?

- a) Hive will create the partition automatically and insert the data.
- b) The data will be discarded, and an error will be raised.
- c) Hive will prompt you to create the partition before inserting the data.
- d) The data will be inserted into the default partition.

Answer: a) Hive will create the partition automatically and insert the data.

Question:

Which of the following statements is true about the "IN" operator in Hive QL?

- a) The "IN" operator can only be used with numerical data types.
- b) The "IN" operator is used to perform pattern matching in Hive.
- c) The "IN" operator can be used to check whether a value exists within a specified list of values.
- d) The "IN" operator is used to perform arithmetic calculations in Hive.

Answer: c) The "IN" operator can be used to check whether a value exists within a specified list of values.

Question:

What is the purpose of the "SORT BY" clause in Hive QL?

- a) It is used to sort the query results in ascending order based on the specified column(s).
- b) The "SORT BY" clause is used to perform a distributed sort across the cluster.
- c) It is used to sort the query results in descending order based on the specified column(s).
- d) The "SORT BY" clause is used to filter rows based on a specific condition.

Answer: a) It is used to sort the query results in ascending order based on the specified column(s).

Question:

Which of the following statements about Hive QL joins is correct?

- a) Hive QL only supports INNER JOIN; other join types are not supported.
- b) Hive QL supports all types of joins, but the syntax is different from standard SQL.
- c) Hive QL supports only LEFT OUTER JOIN and RIGHT OUTER JOIN.
- d) Hive QL joins are not supported; only subqueries can be used for data merging.

Answer: b) Hive QL supports all types of joins, but the syntax is different from standard SQL.

Question:

What does the "DISTRIBUTE BY" clause do in Hive QL?

- a) It is used to specify the distribution method for the data within a partition.
- b) The "DISTRIBUTE BY" clause ensures that the query runs in parallel across all nodes in the cluster.
- c) It is used to specify the order in which data is read from a table.
- d) The "DISTRIBUTE BY" clause is used to perform a distributed group-by operation.

Answer: a) It is used to specify the distribution method for the data within a partition.

Question:

Which Hive QL clause is used to limit the number of rows returned by a query?

- a) LIMIT
- b) ROWS
- c) TOP
- d) FETCH

Answer: a) LIMIT

Question:

In Hive QL, what is the purpose of the "LATERAL VIEW" keyword?

- a) It is used to perform joins on multiple tables with complex relationships.
- b) The "LATERAL VIEW" is used to pivot data in a table and convert rows into columns.
- c) It is used to unnest the elements of an array or map and generate multiple rows for each element.
- d) The "LATERAL VIEW" keyword is used to execute subqueries and derive intermediate results.

Answer: c) It is used to unnest the elements of an array or map and generate multiple rows for each element.

What is Apache Spark's primary programming abstraction for handling large-scale data processing?

- a) MapReduce
- b) DataFrames
- c) SparkSQL
- d) RDDs

Answer: d) RDDs

Which Spark component enables real-time stream processing?

- a) Spark MLlib
- b) Spark Streaming
- c) Spark GraphX
- d) Spark Structured Streaming

Answer: d) Spark Structured Streaming

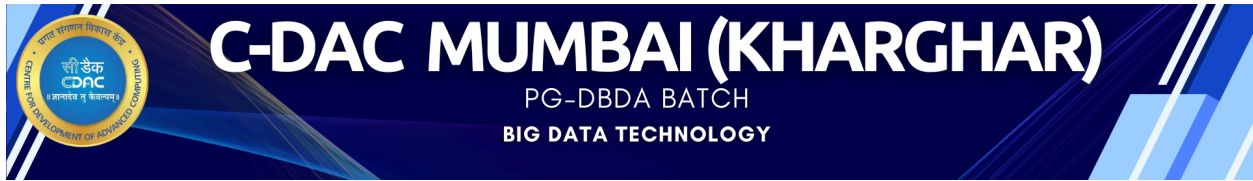
What is the default micro-batch interval in Spark Structured Streaming?

- a) 1 second
- b) 2 seconds
- c) 5 seconds
- d) 10 seconds

Answer: c) 5 seconds

Which Spark function allows you to apply a transformation on each element of an RDD?

- a) reduce
- b) map



- c) filter
- d) collect

Answer: b) map

In Spark Streaming, what does "windowing" refer to?

- a) Dividing the input data into fixed-size batches
- b) Sliding the window over the data stream to perform computations on a set of data points
- c) Aggregating data into a single output at the end of each micro-batch
- d) Dynamically resizing the Spark cluster based on the data load

Answer: b) Sliding the window over the data stream to perform computations on a set of data points

What is the primary advantage of using Spark Structured Streaming over Spark Streaming?

- a) Lower latency
- b) Higher throughput
- c) Simplified API with DataFrame operations
- d) Support for micro-batch processing

Answer: c) Simplified API with DataFrame operations

Which of the following is NOT a supported data source for Spark Structured Streaming?

- a) Kafka
- b) HDFS
- c) Amazon S3
- d) MySQL

Answer: d) MySQL

What is "watermarking" in Spark Structured Streaming?

- a) Adding timestamps to the data stream for synchronization
- b) Cleaning up outdated stateful data in the stream
- c) Resolving conflicts in micro-batch processing
- d) Adjusting the parallelism of the Spark cluster

Answer: b) Cleaning up outdated stateful data in the stream

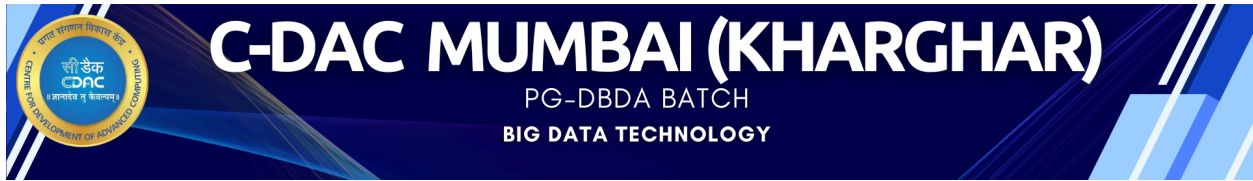
Which Spark operation is used to count the number of elements in an RDD?

- a) countByKey
- b) count
- c) countByValue
- d) length

Answer: b) count

What is the primary benefit of using a stateful transformation in Spark Structured Streaming?

- a) Reduced memory usage
- b) Improved fault tolerance



- c) Better performance
 - d) Ability to maintain state across micro-batches
- Answer: d) Ability to maintain state across micro-batches

Which Spark transformation allows you to combine the elements of an RDD into a single value?

- a) fold
- b) reduceByKey
- c) collect
- d) aggregate

Answer: a) fold

In Spark, what is the purpose of the "checkpoint" directory?

- a) Storing intermediate data during a Spark job
- b) Enabling fault-tolerance in Spark Streaming
- c) Caching the data for faster processing
- d) Identifying the Spark cluster leader node

Answer: b) Enabling fault-tolerance in Spark Streaming

Which Spark deployment mode allows multiple applications to share the same cluster?

- a) Client mode
- b) Cluster mode
- c) Local mode
- d) Standalone mode

Answer: d) Standalone mode

Which of the following is true about "Spark Executors"?

- a) Executors are the driver program in Spark.
- b) Executors are responsible for scheduling tasks on the cluster.
- c) Executors are responsible for executing tasks in parallel.
- d) Executors store the metadata of the Spark application.

Answer: c) Executors are responsible for executing tasks in parallel.

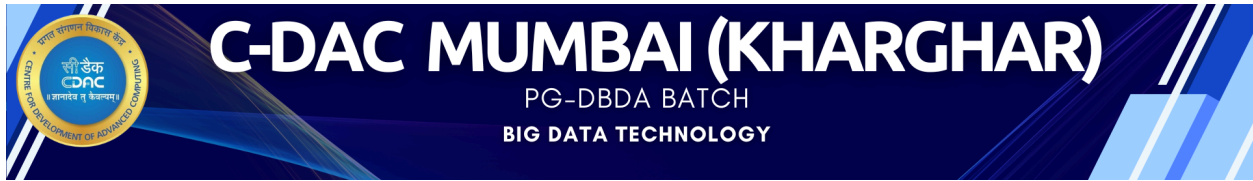
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Which Spark transformation allows you to filter elements of an RDD based on a condition?

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Answer: c) filter

What is the default serialization format used by Spark for transferring data between nodes in a cluster?

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- c) Controlling the parallelism level of a Spark application.
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Answer: b) Summing up values across Spark tasks in a fault-tolerant manner.

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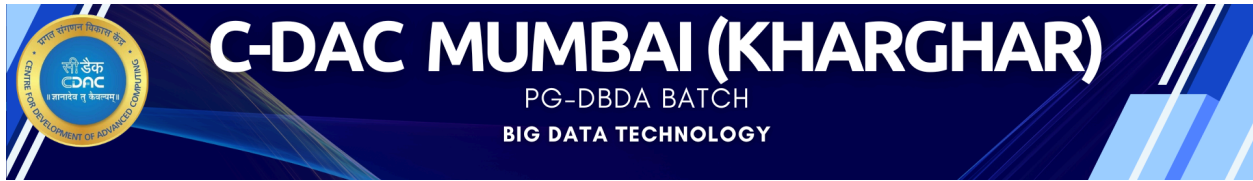
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Answer: c) By using watermarking and checkpointing.



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- d) Fine-grained operations have a higher level of parallelism than coarse-grained operations.

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What is the role of the "SparkContext" object in a Spark application?

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- c) Providing access to Spark functionality and resources.
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Answer: c) Providing access to Spark functionality and resources.

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- b) Kafka
- c) Amazon S3
- d) MongoDB

Answer: d) MongoDB

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- a) To perform aggregation over a sliding window of data.
- b) To filter data based on a specific condition.
- c) To join two DataFrames based on a common key.
- d) To create a new DataFrame by combining multiple DataFrames.

Answer: a) To perform aggregation over a sliding window of data.

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- c) By configuring a large number of Spark Executors to increase parallelism.
- d) By reducing the number of partitions in the input data.

Answer: c) By configuring a large number of Spark Executors to increase parallelism.

What does the "checkpoint directory" in Spark Streaming allow you to do?

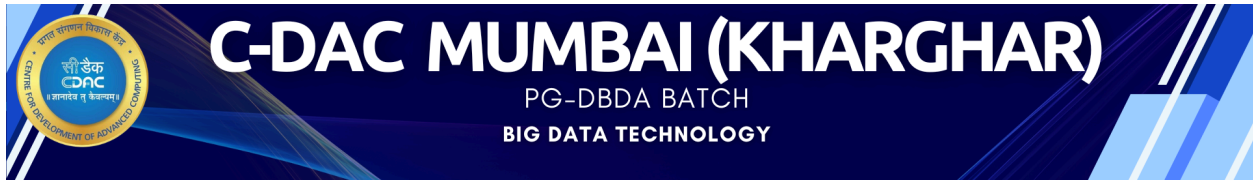
- a) To store the final results of a Spark job.
- b) To store intermediate data during iterative processing.
- c) To enable fault-tolerance by persisting the metadata of the streaming application.
- d) To schedule periodic checkpoints to improve performance.

Answer: c) To enable fault-tolerance by persisting the metadata of the streaming application.

Which of the following is NOT a Spark Streaming source that can be used to ingest data?

- a) Kafka
- b) Amazon S3
- c) Flume
- d) HDFS

Answer: d) HDFS



In Spark, what is the function of the "cache" and "persist" operations?

- a) To store data in memory for faster access.
- b) To compress data for efficient storage.
- c) To shuffle data across the cluster for parallel processing.
- d) To checkpoint data for fault tolerance.

Answer: a) To store data in memory for faster access.

Which Spark operation is used to compute the sum of values in an RDD?

- a) sum
- b) reduce
- c) aggregate
- d) fold

Answer: b) reduce

What is the primary benefit of using Spark Streaming instead of traditional batch processing?

- a) Spark Streaming allows you to process data in real-time.
- b) Spark Streaming offers better fault-tolerance capabilities.
- c) Spark Streaming guarantees exactly-once processing semantics.
- d) Spark Streaming supports a wider variety of data sources.

Answer: a) Spark Streaming allows you to process data in real-time.

What is the function of the "transform" operation in Spark Structured Streaming?

- a) To map the elements of an RDD to a new RDD.
- b) To perform data aggregation over a window of time.
- c) To apply custom, non-aggregate logic on the DataFrames.
- d) To join two DataFrames based on a common key.

Answer: c) To apply custom, non-aggregate logic on the DataFrames.

How does Spark Structured Streaming handle schema evolution when processing data streams?

- a) It automatically adapts to schema changes without any manual intervention.
- b) It requires you to manually update the schema before processing new data.
- c) It automatically applies the same schema used during the initial processing to all subsequent data.
- d) It rejects any data with schema changes to maintain consistency.

Answer: a) It automatically adapts to schema changes without any manual intervention.

In Spark Structured Streaming, what is the role of the "output mode"?

- a) To control how the final output is written to external storage.
- b) To specify the data format of the input sources.
- c) To define the windowing interval for processing data.
- d) To determine the order of data processing in the streaming application.

Answer: a) To control how the final output is written to external storage.

How does Spark handle data serialization and deserialization?

- a) Spark uses Java Object Serialization by default.
- b) Spark automatically selects the most efficient serialization format based on the data type.
- c) Users can configure the serialization format manually.
- d) Spark always uses Avro for serialization and deserialization.

Answer: a) Spark uses Java Object Serialization by default.

Which Spark operation is used to find the maximum value in an RDD?

- a) max
- b) maximum
- c) reduce
- d) collect

Answer: a) max

What is the role of the "executor memory" configuration in Spark?

- a) It controls the amount of memory allocated to Spark Executors.
- b) It specifies the amount of memory allocated to the Spark Driver program.
- c) It controls the amount of memory allocated to the cluster manager.
- d) It determines the total memory available for the Spark application.

Answer: a) It controls the amount of memory allocated to Spark Executors.

What does "RDD lineage" refer to in Spark?

- a) The transformation of an RDD into a DataFrame.
 - b) The sequence of transformations applied to an RDD to calculate its data.
 - c) The mapping of an RDD to the physical memory of Spark Executors.
 - d) The replication of an RDD across multiple nodes in the cluster.
- Answer: b) The sequence of transformations applied to an RDD to calculate its data.

Which Spark operation is used to perform element-wise addition of two RDDs?

- a) add
- b) plus
- c) union
- d) zip

Answer: d) zip

Question:

In Hive QL, which type of table supports ACID (Atomicity, Consistency, Isolation, Durability) properties?

- a) Managed table
- b) External table
- c) Transactional table

d) Temporary table

Answer: c) Transactional table

Question:

What is the purpose of the Hive QL "CLUSTERED BY" clause?

- a) It specifies the number of clusters to be used in a data warehouse.
- b) The "CLUSTERED BY" clause ensures that all rows with the same key value are stored together in the same partition.
- c) It is used to physically organize the data within a table based on a specific column or columns.
- d) The "CLUSTERED BY" clause is used to group rows based on a specific condition during query execution.

Answer: c) It is used to physically organize the data within a table based on a specific column or columns.

Question:

Which of the following Hive QL functions is used to convert a string to lowercase?

- a) LOWER()
- b) TOLOWER()
- c) LOWERCASE()
- d) LCASE()

Answer: a) LOWER()

Question:

What will be the result of the following Hive QL query?

sql

Copy code

```
SELECT count(DISTINCT department) FROM employees;
```

- a) It counts the total number of employees in each department.
- b) The query returns an error since "DISTINCT" is not allowed with the "COUNT" function.
- c) It counts the number of unique departments in the "employees" table.
- d) The query returns the sum of the "department" column for each employee.

Answer: c) It counts the number of unique departments in the "employees" table.

Question:

In Hive QL, what does the "ANALYZE TABLE" command do?

- a) It generates a report showing the table's schema and metadata.
- b) The "ANALYZE TABLE" command is used to calculate statistics for a table to improve query optimization.
- c) It creates a backup of the table data in case of accidental data loss.
- d) The "ANALYZE TABLE" command is used to rename a table in Hive.

Answer: b) The "ANALYZE TABLE" command is used to calculate statistics for a table to improve query optimization.

Question:

What does the Hive QL "CONCAT_WS" function do?

- a) It concatenates two strings with a specified separator.
- b) The "CONCAT_WS" function is used to concatenate all columns of a table into a single string.
- c) It converts a string to a timestamp by appending a default time.
- d) The "CONCAT_WS" function is used to concatenate two columns with a specified separator.

Answer: a) It concatenates two strings with a specified separator.

Question:

What happens when you run the following Hive QL query?

sql

Copy code

```
INSERT OVERWRITE TABLE target_table PARTITION (date='2023-07-01') SELECT * FROM source_table;
```

- a) The query appends data from "source_table" into "target_table" for the partition date '2023-07-01'.
- b) The query replaces the data in "target_table" for the partition date '2023-07-01' with data from "source_table."
- c) The query raises an error because partition-specific inserts are not allowed in Hive QL.
- d) The query appends data from "source_table" into "target_table" without considering any partitioning.

Answer: b) The query replaces the data in "target_table" for the partition date '2023-07-01' with data from "source_table."

Question:

What is the purpose of the Hive QL "TRANSFORM" clause?

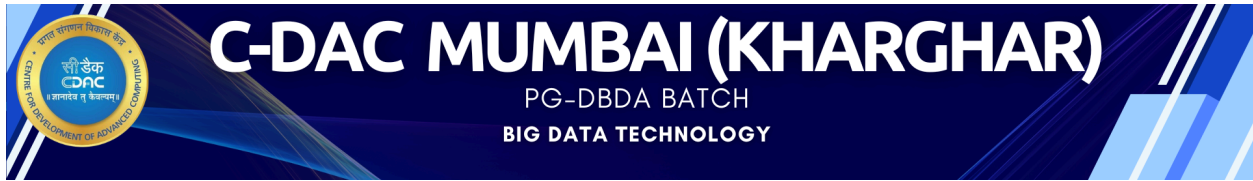
- a) It is used to perform complex mathematical transformations on numeric data.
- b) The "TRANSFORM" clause is used to create a temporary transformation function for use within a query.
- c) It enables data transformation while loading data into a Hive table from an external source.
- d) The "TRANSFORM" clause is used to execute a user-defined script or executable on each input row.

Answer: d) The "TRANSFORM" clause is used to execute a user-defined script or executable on each input row

Question:

In Hive QL, what is the purpose of the "LATERAL VIEW OUTER" keyword in conjunction with the "EXPLODE" function?

- a) It allows you to perform an outer join with a subquery using the "EXPLODE" function.
- b) The "LATERAL VIEW OUTER" keyword is used to unnest arrays or maps, and "EXPLODE" handles unnesting for NULL values.
- c) It is used to explode arrays or maps into multiple rows, and "OUTER" specifies that NULL values should be included.



d) The "LATERAL VIEW OUTER" keyword enables you to apply a user-defined function to each element of an array or map using "EXPLODE."

Answer: c) It is used to explode arrays or maps into multiple rows, and "OUTER" specifies that NULL values should be included.

Question:

Which of the following Hive QL functions is used to return the current date and time?

- a) GETDATE()
- b) NOW()
- c) CURRENT_TIMESTAMP()
- d) TODAY()

Answer: c) CURRENT_TIMESTAMP()

Question:

In Hive QL, what will happen if you try to create a table with a name that contains spaces?

- a) Hive QL automatically replaces the spaces with underscores in the table name.
- b) The table creation will fail, as table names cannot contain spaces in Hive.
- c) Hive QL prompts you to enter an alternative name without spaces.
- d) The table name will be accepted, but it can cause compatibility issues with certain Hive clients.

Answer: b) The table creation will fail, as table names cannot contain spaces in Hive.

Question:

Which Hive QL statement should be used to view the schema and structure of a table named "employees"?

- a) DESCRIBE TABLE employees;
- b) SHOW SCHEMA employees;
- c) EXPLAIN employees;
- d) DISPLAY TABLE employees;

Answer: a) DESCRIBE TABLE employees;

Question:

What is the purpose of the Hive QL "CREATE VIEW" statement?

- a) It creates a physical table with the data specified in the query.
- b) The "CREATE VIEW" statement is used to define a virtual table that is based on the result of a query.
- c) It creates a temporary table that persists only for the duration of the Hive session.
- d) The "CREATE VIEW" statement is used to define a table schema without any data.

Answer: b) The "CREATE VIEW" statement is used to define a virtual table that is based on the result of a query.

Question:

Which Hive QL JOIN type returns all rows from both tables and combines rows when there is a match?

- a) LEFT JOIN

b) RIGHT JOIN

c) FULL OUTER JOIN

d) INNER JOIN

Answer: c) FULL OUTER JOIN

Question:

What does the Hive QL "ORDER BY" clause do when used in conjunction with the "CLUSTERED BY" clause?

a) It specifies the order in which the data is stored within each cluster in a Hive table.

b) The "ORDER BY" clause ensures that the query results are sorted based on the "CLUSTERED BY" column.

c) It is used to sort the data within each cluster and distribute the sorted data across the cluster nodes.

d) The "ORDER BY" clause is not allowed when using the "CLUSTERED BY" clause in Hive QL.

Answer: d) The "ORDER BY" clause is not allowed when using the "CLUSTERED BY" clause in Hive QL.

Question:

In Hive QL, which function is used to find the maximum value from a column?

a) MAX()

b) GREATEST()

c) LARGEST()

d) TOP()

Answer: a) MAX()

Question:

In Hive QL, what is the purpose of the "DISTRIBUTE BY" clause when used with the "SORT BY" clause in the same query?

a) It ensures that the data is distributed evenly across all nodes in the cluster.

b) The "DISTRIBUTE BY" clause defines the sort order, while the "SORT BY" clause defines the distribution order.

c) It is used to specify the columns on which data should be sorted before distribution.

d) The "DISTRIBUTE BY" clause is not required when using the "SORT BY" clause in Hive QL.

Answer: a) It ensures that the data is distributed evenly across all nodes in the cluster.

Question:

Which Hive QL keyword is used to create a new partition for an existing table?

a) ADD PARTITION

b) CREATE PARTITION

c) INSERT PARTITION

d) PARTITION BY

Answer: a) ADD PARTITION

Question:



What is the purpose of the "PERSISTENT" keyword when creating a Hive QL function?

- a) It ensures that the function's results are stored permanently in HDFS.
- b) The "PERSISTENT" keyword is used to define a function that can be called from any Hive query.
- c) It specifies that the function's data is not to be cached in memory for better performance.
- d) The "PERSISTENT" keyword is not used in Hive QL function creation.

Answer: d) The "PERSISTENT" keyword is not used in Hive QL function creation.

Question:

In Hive QL, which function is used to calculate the cumulative distribution function (CDF) of a numerical column?

- a) CUME_DIST()
- b) PERCENT_RANK()
- c) RANK()
- d) PERCENTILE()

Answer: a) CUME_DIST()

Question:

What happens when you run the following Hive QL query?

```
SET hive.optimize.skewjoin=true;
```

- a) It enables the optimization for skew joins, which helps balance the data distribution during joins.
- b) The query raises an error since "hive.optimize.skewjoin" is not a valid configuration parameter.
- c) It disables the optimization for skew joins, which reduces query execution time.
- d) The query changes the Hive QL optimization level to "true" for all subsequent queries.

Answer: a) It enables the optimization for skew joins, which helps balance the data distribution during joins.

Question:

Which Hive QL keyword is used to create a copy of an existing table?

- a) CREATE
- b) COPY
- c) DUPLICATE
- d) LIKE

Answer: d) LIKE

Question:

What is the purpose of the "CONCAT" function in Hive QL?

- a) It is used to concatenate two or more strings together.
- b) The "CONCAT" function is used to merge two or more tables into a single table.
- c) It calculates the concatenation of all values in a numeric column.
- d) The "CONCAT" function converts numeric values to strings.

Answer: a) It is used to concatenate two or more strings together.

Question:

In Hive QL, what will happen if you try to insert data into a partitioned table without specifying the partition columns?

- a) The data will be inserted into the first partition of the table.
- b) The query will fail as Hive QL requires explicit partition column values for insertion.
- c) The data will be inserted into all partitions of the table in round-robin fashion.
- d) Hive will prompt you to create the necessary partitions before inserting the data.

Answer: b) The query will fail as Hive QL requires explicit partition column values for insertion

In Hive QL, which statement is true about the "LIMIT" clause in a SELECT query?

- a) The "LIMIT" clause must always be the last clause in a SELECT query.
- b) The "LIMIT" clause can only be used with numerical columns.
- c) The "LIMIT" clause is used to filter rows based on a specific condition.
- d) The "LIMIT" clause restricts the number of rows returned by the query.

Answer: d) The "LIMIT" clause restricts the number of rows returned by the query.

Question:

What is the purpose of the Hive QL "PARTITIONED BY" clause when creating a table?

- a) It defines the columns that will be indexed for faster retrieval.
- b) The "PARTITIONED BY" clause specifies the data type of the partition columns.
- c) It is used to split the data into separate directories based on the partitioning columns.
- d) The "PARTITIONED BY" clause enables distributed data storage across all nodes in the cluster.

Answer: c) It is used to split the data into separate directories based on the partitioning columns.

Question:

In Hive QL, which of the following aggregate functions can be used with non-numeric data types?

- a) AVG()
- b) COUNT()
- c) SUM()
- d) MAX()

Answer: b) COUNT()

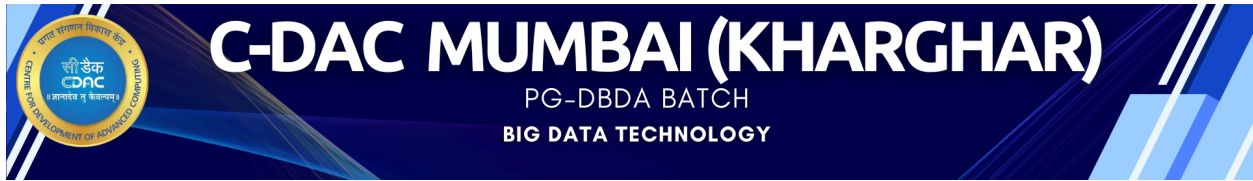
Question:

What happens when you execute the Hive QL command "TRUNCATE TABLE employees;"?

- a) The table schema is deleted, but the data remains intact.
- b) It permanently removes all data and the table schema.
- c) The table is dropped, and all data is deleted permanently.
- d) The command raises an error, as "TRUNCATE" is not supported in Hive QL.

Answer: b) It permanently removes all data and the table schema.

Question:



In Hive QL, what does the "INSERT OVERWRITE DIRECTORY" statement do?

- a) It inserts data into a new directory on HDFS, creating a new table in the process.
- b) The "INSERT OVERWRITE DIRECTORY" statement is used to replace existing data in a directory with new data.
- c) It overwrites the data in an existing Hive table with data from another table or query result.
- d) The "INSERT OVERWRITE DIRECTORY" statement is not supported in Hive QL.

Answer: b) The "INSERT OVERWRITE DIRECTORY" statement is used to replace existing data in a directory with new data.

Question:

Which Hive QL JOIN type returns only the rows that have matching values in both tables?

- a) LEFT JOIN
- b) RIGHT JOIN
- c) FULL OUTER JOIN
- d) INNER JOIN

Answer: d) INNER JOIN

Question:

What is the purpose of the "STORED AS" clause in Hive QL when creating a table?

- a) It specifies the file format to be used for storing the table data on HDFS.
- b) The "STORED AS" clause defines the storage properties for the table, such as compression and encryption.
- c) It is used to specify the default partitioning scheme for the table.
- d) The "STORED AS" clause indicates whether the table data should be stored in a managed or external table.

Answer: a) It specifies the file format to be used for storing the table data on HDFS.

Question:

In Hive QL, what is the purpose of the "UNION" keyword?

- a) It combines the results of two or more SELECT queries into a single result set, removing duplicates.
- b) The "UNION" keyword is used to merge two or more tables with similar schema into a new table.
- c) It performs a set operation that returns only the common rows between two SELECT queries.
- d) The "UNION" keyword is not supported in Hive QL.

Answer: a) It combines the results of two or more SELECT queries into a single result set, removing duplicates.

Question:

Which of the following statements about Hive QL functions is true?

- a) Hive QL functions cannot be used in subqueries.
- b) User-defined functions (UDFs) are the only type of functions supported in Hive QL.
- c) Hive QL functions can only be used in the SELECT clause of a query.
- d) Hive QL provides a wide range of built-in functions for various data types and operations.

Answer: d) Hive QL provides a wide range of built-in functions for various data types and operations.

Question:

What is the purpose of the "NVL" function in Hive QL?

- a) It converts a NULL value to a specified default value.
- b) The "NVL" function is used to extract the numeric value of a string.
- c) It calculates the variance of a numeric column.
- d) The "NVL" function is used to return the index of a specific element in an array.

Answer: a) It converts a NULL value to a specified default value.

Question:

In Hive QL, which statement is true about bucketing?

- a) Bucketing is a technique used to group data into smaller, evenly sized partitions based on a specified column.
- b) The number of buckets used in a table should always be equal to the number of data nodes in the cluster.
- c) Bucketing is only applicable when using the ORC file format in Hive.
- d) The bucketing column must always be a partitioning column.

Answer: a) Bucketing is a technique used to group data into smaller, evenly sized partitions based on a specified column.

Question:

What is the purpose of the "LATERAL VIEW OUTER EXPLODE" clause in Hive QL?

- a) It performs an outer join with a subquery using the "EXPLODE" function.
- b) The "LATERAL VIEW OUTER EXPLODE" clause is used to unnest arrays or maps, and "EXPLODE" handles unnesting for NULL values.
- c) It is used to explode arrays or maps into multiple rows, and "OUTER" specifies that NULL values should be included.
- d) The "LATERAL VIEW OUTER EXPLODE" clause enables you to apply a user-defined function to each element of an array or map using "EXPLODE."

Answer: c) It is used to explode arrays or maps into multiple rows, and "OUTER" specifies that NULL values should be included.

Question:

In Hive QL, what will happen if you use the "SORT BY" clause without the "DISTRIBUTE BY" clause in the same query?

- a) The query will raise an error, as "SORT BY" must always be used with "DISTRIBUTE BY."
- b) The "SORT BY" clause will not have any effect on the query results without "DISTRIBUTE BY."
- c) It will sort the data within each reducer without affecting the distribution across the cluster.
- d) The query will sort the data globally across all nodes in the cluster.

Answer: c) It will sort the data within each reducer without affecting the distribution across the cluster.



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Question:

What is the purpose of the "STREAMTABLE" keyword in Hive QL?

- a) It is used to create a temporary table that holds the output of a streaming query.
- b) The "STREAMTABLE" keyword is used to enable real-time data processing in Hive QL.
- c) It converts a regular table into a streaming table that continuously updates data.
- d) The "STREAMTABLE" keyword is not a valid keyword in Hive QL.

Answer: d) The "STREAMTABLE" keyword is not a valid keyword in Hive QL.

Question:

Which Hive QL function is used to calculate the mean of a numeric column?

- a) AVG()
- b) MEAN()
- c) SUM()
- d) AVERAGE()

Answer: a) AVG()

Question:

What will happen if you execute the following Hive QL command?

SET hive.exec.dynamic.partition=true;

- a) It enables dynamic partitioning for all subsequent Hive QL queries in the current session.
- b) The query raises an error, as "hive.exec.dynamic.partition" is not a valid configuration parameter.
- c) It disables dynamic partitioning for all subsequent Hive QL queries in the current session.
- d) The command changes the Hive QL execution mode to "dynamic partitioning" for all queries.

Answer: a) It enables dynamic partitioning for all subsequent Hive QL queries in the current session.

Question:

In Hive QL, which function is used to calculate the percentile of a numeric column?

- a) PERCENT_RANK()
- b) PERCENTILE()
- c) PERCENTILE_APPROX()
- d) PERCENTILE_DISC()

Answer: c) PERCENTILE_APPROX()

Question:

What is the purpose of the "USING" keyword in the Hive QL "INSERT INTO" statement?

- a) It specifies the data format of the inserted data, such as JSON or XML.
- b) The "USING" keyword is used to indicate the data source from which data is to be inserted.
- c) It defines the join condition between the source and target tables during the insert operation.
- d) The "USING" keyword is not used in the Hive QL "INSERT INTO" statement.

Answer: d) The "USING" keyword is not used in the Hive QL "INSERT INTO" statement.



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Question:

In Hive QL, what is the purpose of the "CACHE" keyword?

- a) It is used to cache the query result in memory for faster access during subsequent queries.
- b) The "CACHE" keyword is used to store the query result in a separate table for future use.
- c) It is used to cache the query plan for the current Hive QL query.
- d) The "CACHE" keyword is not supported in Hive QL.

Answer: d) The "CACHE" keyword is not supported in Hive QL.

Question:

What does the Hive QL "CLUSTERED BY" clause do when used in conjunction with the "SORTED BY" clause in the same query?

- a) It ensures that the data is distributed evenly across all nodes in the cluster.
- b) The "CLUSTERED BY" clause defines the sort order, while the "SORTED BY" clause defines the distribution order.
- c) It is used to specify the columns on which data should be sorted before distribution.
- d) The "CLUSTERED BY" clause is not allowed when using the "SORTED BY" clause in Hive QL.

Answer: b) The "CLUSTERED BY" clause defines the sort order, while the "SORTED BY" clause defines the distribution order.

Question:

Which Hive QL JOIN type returns all rows from the left table and only the matching rows from the right table?

- a) LEFT JOIN
- b) RIGHT JOIN
- c) FULL OUTER JOIN
- d) INNER JOIN

Answer: a) LEFT JOIN

Question:

What is the purpose of the "TRANSFORM" clause in Hive QL?

- a) It is used to perform complex mathematical transformations on numeric data.
- b) The "TRANSFORM" clause is used to create a temporary transformation function for use within a query.
- c) It enables data transformation while loading data into a Hive table from an external source.
- d) The "TRANSFORM" clause is used to execute a user-defined script or executable on each input row.

Answer: d) The "TRANSFORM" clause is used to execute a user-defined script or executable on each input row.

Question:

In Hive QL, what does the "ORDER BY" clause do when used with the "DISTRIBUTE BY" clause in the same query?

- a) It specifies the order in which data is stored within each cluster in a Hive table.

b) The "ORDER BY" clause ensures that the query results are sorted based on the "DISTRIBUTE BY" column.

c) It is used to sort the data within each cluster and distribute the sorted data across the cluster nodes.

d) The "ORDER BY" clause is not allowed when using the "DISTRIBUTE BY" clause in Hive QL.

Answer: b) The "ORDER BY" clause ensures that the query results are sorted based on the "DISTRIBUTE BY" column.

Question:

What is the purpose of the "TRANSFORM" clause when used with the "USING" keyword in Hive QL?

a) It specifies the data format of the external file to be used in the transformation.

b) The "TRANSFORM" clause is used to define a custom transformation function in Hive QL.

c) It is used to execute a custom transformation script on each row of the external data.

d) The "TRANSFORM" clause is not used with the "USING" keyword in Hive QL.

Answer: c) It is used to execute a custom transformation script on each row of the external data.

Question:

In Hive QL, which function is used to calculate the rank of each row in a result set based on a specified column?

a) RANK()

b) ROW_NUMBER()

c) DENSE_RANK()

d) ORDER()

Answer: b) ROW_NUMBER()

In Hive QL, which keyword is used to create a new database?

a) CREATE

b) DATABASE

c) NEW

d) SCHEMA

Answer: b) DATABASE

Question:

What will be the output of the following Hive QL query?

sql

Copy code

```
SELECT COUNT(*) FROM employees;
```

a) The total number of employees in the "employees" table.

b) The number of columns in the "employees" table.

c) The number of distinct values in the "employees" table.

d) An error, as "COUNT(*)" is not allowed in Hive QL.

Answer: a) The total number of employees in the "employees" table.

Question:

Which Hive QL function is used to calculate the standard deviation of a numeric column?

- a) VARIANCE()
- b) AVG()
- c) STDDEV()
- d) STDEV()

Answer: c) STDDEV()

Question:

What happens when you execute the following Hive QL command?

```
sql
```

```
Copy code
```

```
SET hive.execution.engine=mr;
```

- a) It switches the Hive execution engine to Tez for all subsequent queries in the current session.
- b) The query raises an error, as "hive.execution.engine" is not a valid configuration parameter.
- c) It enables the use of the MapReduce execution engine for all subsequent queries in the current session.
- d) The command changes the Hive execution engine to Spark for all queries.

Answer: c) It enables the use of the MapReduce execution engine for all subsequent queries in the current session.

Question:

In Hive QL, which keyword is used to load data from an HDFS file into a table?

- a) INSERT INTO
- b) LOAD
- c) INGEST
- d) PUT

Answer: b) LOAD

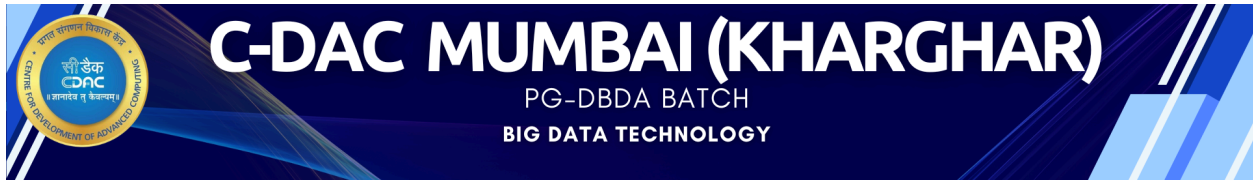
Question:

What is the purpose of the Hive QL "LATERAL VIEW OUTER INLINE" clause?

- a) It performs an outer join with a subquery using the "INLINE" function.
- b) The "LATERAL VIEW OUTER INLINE" clause is used to unnest arrays or maps, including NULL values.
- c) It is used to explode arrays or maps into multiple rows, including NULL values.
- d) The "LATERAL VIEW OUTER INLINE" clause enables you to apply a user-defined function to each element of an array or map using "INLINE."

Answer: b) The "LATERAL VIEW OUTER INLINE" clause is used to unnest arrays or maps, including NULL values.

Question:



What is the purpose of the "IF" function in Hive QL?

- a) It evaluates a condition and returns one value if the condition is true, and another value if the condition is false.
- b) The "IF" function is used to check if a column contains NULL values.
- c) It calculates the average of numeric values in a column.
- d) The "IF" function is not supported in Hive QL.

Answer: a) It evaluates a condition and returns one value if the condition is true, and another value if the condition is false.

Question:

What will be the result of the following Hive QL query?

sql

Copy code

```
SELECT department, COUNT(*) FROM employees GROUP BY department;
```

- a) It returns the count of employees in each department.
- b) The query raises an error because "COUNT(*)" cannot be used with "GROUP BY" in Hive QL.
- c) It calculates the average count of employees across all departments.
- d) The query counts the total number of departments in the "employees" table.

Answer: a) It returns the count of employees in each department.

Question:

In Hive QL, which statement is true about the "ORC" file format?

- a) The ORC file format is the default format used in Hive QL for all tables.
- b) ORC is a compressed file format optimized for fast reads and smaller storage.
- c) The ORC file format is only suitable for text-based data.
- d) ORC files can be created in Hive QL using the "STORED AS ORC" clause.

Answer: b) ORC is a compressed file format optimized for fast reads and smaller storage.

Question:

What does the Hive QL "DROP TABLE IF EXISTS" statement do?

- a) It drops the table specified and all its data without any confirmation.
- b) The "DROP TABLE IF EXISTS" statement removes all partitions from the table.
- c) It is used to truncate the table data without dropping the table schema.
- d) The "DROP TABLE IF EXISTS" statement is not supported in Hive QL.

Answer: a) It drops the table specified and all its data without any confirmation.

Question:

In Hive QL, which function is used to convert a timestamp to a string representation?

- a) `TIMESTAMP_TO_STRING()`
- b) `TO_STRING()`
- c) `FROM_TIMESTAMP()`

d) DATE_FORMAT()

Answer: d) DATE_FORMAT()

Question:

What happens when you execute the following Hive QL command?

SET hive.exec.dynamic.partition.mode=nonstrict;

- a) It enables strict mode for dynamic partitioning in Hive QL for all subsequent queries.
- b) The query raises an error, as "hive.exec.dynamic.partition.mode" is not a valid configuration parameter.
- c) It disables dynamic partitioning for all subsequent Hive QL queries in the current session.
- d) The command changes the Hive QL dynamic partitioning mode to non-strict for all queries.

Answer: d) The command changes the Hive QL dynamic partitioning mode to non-strict for all queries.

Question:

In Hive QL, which keyword is used to define a table based on an existing table's schema without copying data?

- a) LIKE
- b) CLONE
- c) AS
- d) FOLLOW

Answer: a) LIKE

Question:

What is the purpose of the "TRANSFORM" clause when used with the "USING 'python script'" keyword in Hive QL?

- a) It specifies the data format of the external file to be used in the transformation.
- b) The "TRANSFORM" clause is used to define a custom transformation function written in Python.
- c) It is used to execute a custom Python script on each row of the external data.
- d) The "TRANSFORM" clause is not used with the "USING 'python script'" keyword in Hive QL.

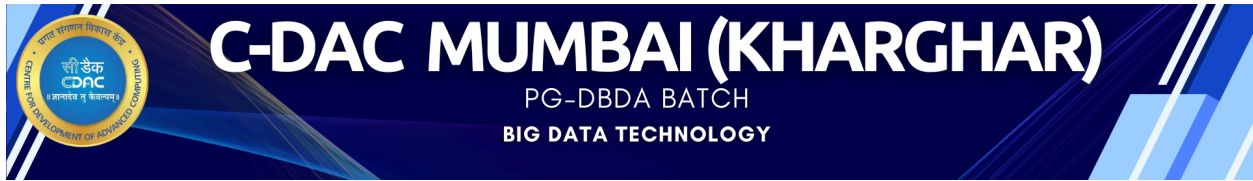
Answer: c) It is used to execute a custom Python script on each row of the external data.

Question:

What will be the output of the following Hive QL query?

SELECT department, salary FROM employees ORDER BY department, salary DESC;

- a) The data in the "employees" table will be sorted by department in ascending order, and salaries within each department will be sorted in descending order.
- b) The data in the "employees" table will be sorted by department in descending order, and salaries within each department will be sorted in ascending order.
- c) The query will raise an error since you cannot use "DESC" with "ORDER BY" in Hive QL.
- d) The query will return the data in the "employees" table without any sorting.



Answer: a) The data in the "employees" table will be sorted by department in ascending order, and salaries within each department will be sorted

Which Hive QL command is used to create a new database?

- a) CREATE SCHEMA db_name;
- b) CREATE DATABASE db_name;
- c) NEW DATABASE db_name;
- d) ADD DATABASE db_name;

Answer: b) CREATE DATABASE db_name;

Question:

What will the following Hive QL query do?

sql

Copy code

```
SELECT name FROM employees WHERE salary BETWEEN 50000 AND 80000;
```

- a) Retrieve the names of employees with a salary between \$50,000 and \$80,000.
- b) Raise an error, as the "BETWEEN" operator is not supported in Hive QL.
- c) Retrieve the names of employees with a salary greater than or equal to \$50,000 and less than or equal to \$80,000.
- d) Retrieve the names of employees with a salary greater than \$50,000 and less than \$80,000.

Answer: c) Retrieve the names of employees with a salary greater than or equal to \$50,000 and less than or equal to \$80,000.

Question:

In Hive QL, what does the following query do?

```
DESCRIBE FORMATTED employees;
```

- a) Returns the detailed schema of the "employees" table.
- b) Lists all the employees' names in the table.
- c) Describes the format of the "employees" table.
- d) Raises an error, as "DESCRIBE FORMATTED" is not a valid Hive QL command.

Answer: a) Returns the detailed schema of the "employees" table.

Question:

Which Hive QL command is used to change the name of a table?

- a) ALTER TABLE table_name RENAME TO new_table_name;
- b) RENAME TABLE table_name TO new_table_name;
- c) CHANGE TABLE table_name new_table_name;
- d) MODIFY TABLE table_name TO new_table_name;

Answer: a) ALTER TABLE table_name RENAME TO new_table_name;

Question:

What is the purpose of the Hive QL "ROW FORMAT DELIMITED FIELDS TERMINATED BY" clause?

- a) It specifies the delimiter used for separating rows in a Hive table.
- b) The "ROW FORMAT DELIMITED" clause is used to define the format of JSON data.
- c) It defines the character used to separate fields within a row of data.
- d) The "ROW FORMAT DELIMITED" clause is not supported in Hive QL.

Answer: c) It defines the character used to separate fields within a row of data.

Question:

What is the result of the following Hive QL query?

sql

Copy code

```
SELECT department, COUNT(*) FROM employees GROUP BY department;
```

- a) Counts the total number of employees in each department.
- b) Retrieves the department names and the count of distinct employees in each department.
- c) Retrieves the department names and the total number of employees in each department.
- d) Raises an error, as "COUNT(*)" cannot be used with "GROUP BY" in Hive QL.

Answer: c) Retrieves the department names and the total number of employees in each department.

Question:

Which Hive QL statement is used to remove all data from a table without deleting the table itself?

- a) DROP DATA FROM table_name;
- b) TRUNCATE TABLE table_name;
- c) DELETE FROM table_name;
- d) CLEAR TABLE table_name;

Answer: b) TRUNCATE TABLE table_name;

Question:

What is the purpose of the Hive QL "EXTERNAL" keyword when creating a table?

- a) It specifies that the table is managed by Hive and its data is stored in the Hive warehouse directory.
- b) The "EXTERNAL" keyword is used to create a table with external data, where Hive manages only the metadata.
- c) It defines that the table has an external partitioning scheme.
- d) The "EXTERNAL" keyword indicates that the table is temporary and will be automatically dropped after the session ends.

Answer: b) The "EXTERNAL" keyword is used to create a table with external data, where Hive manages only the metadata.

Question:



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In Hive QL, what is the purpose of the "SAMPLE" clause in a SELECT query?

- a) It retrieves a random sample of data from the table based on the specified percentage.
- b) The "SAMPLE" clause is used to sort the data in random order before applying the WHERE clause.
- c) It groups the data into samples based on a specified column in the table.
- d) The "SAMPLE" clause is used to filter out NULL values from the query result.

Answer: a) It retrieves a random sample of data from the table based on the specified percentage.

Question:

Which Hive QL statement is used to add a new column to an existing table?

- a) ALTER TABLE table_name ADD COLUMN column_name data_type;
- b) ADD COLUMN column_name data_type TO table_name;
- c) ALTER TABLE table_name MODIFY COLUMN column_name data_type;
- d) CREATE COLUMN column_name data_type FOR table_name;

Answer: a) ALTER TABLE table_name ADD COLUMN column_name data_type;

Question:

What does the following Hive QL query do?

sql

Copy code

```
SELECT DISTINCT department FROM employees;
```

- a) Retrieves all columns for distinct department names from the "employees" table.
- b) Returns the unique department names from the "employees" table.
- c) Retrieves the departments where there are no employees.
- d) Raises an error, as "DISTINCT" cannot be used without an aggregate function.

Answer: b) Returns the unique department names from the "employees" table.

Question:

In Hive QL, what is the purpose of the "CONCAT_WS" function?

- a) It concatenates two strings with a specified separator.
- b) The "CONCAT_WS" function is used to merge data from two different tables.
- c) It converts a string to uppercase.
- d) The "CONCAT_WS" function is used to remove leading and trailing whitespace from a string.

Answer: a) It concatenates two strings with a specified separator.

Question:

What is the purpose of the Hive QL "SHOW TABLES" command?

- a) It displays a list of available databases in Hive.
- b) The "SHOW TABLES" command is used to view the structure of a specific table.
- c) It lists the tables in the current database.
- d) The "SHOW TABLES" command is not supported in Hive QL.

Answer: c) It lists the tables in the current database.

Question:

In Hive QL, which function is used to calculate the difference between two dates in days?

- a) DATEDIFF()
- b) DATE_DIFF()
- c) DAYDIFF()
- d) DIFF_DATE()

Answer: a) DATEDIFF()

Question:

What does the following Hive QL query do?

```
SELECT * FROM employees WHERE name LIKE 'J%';
```

- a) Retrieves all columns for employees whose names start with 'J'.
- b) Returns the employees with the shortest names in the table.
- c) Retrieves the total number of employees whose names contain 'J'.
- d) Raises an error, as the "LIKE" operator cannot be used without an aggregate function.

Answer: a) Retrieves all columns for employees whose names start with 'J'.

Question:

Which Hive QL keyword is used to specify a condition that must be met for a row to be included in a query result?

- a) FILTER
- b) HAVING
- c) WHERE
- d) CONDITION

Answer: c) WHERE

Question:

What is the purpose of the Hive QL "SHOW CREATE TABLE" command?

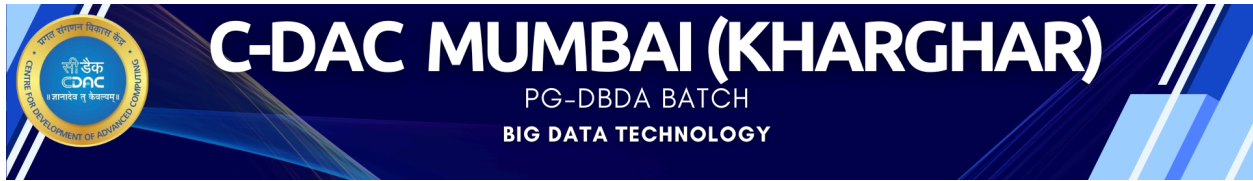
- a) It displays the definition of a specified table, including the table schema and metadata.
- b) The "SHOW CREATE TABLE" command is used to view the structure of a specific database.
- c) It lists all the partitions of a table.
- d) The "SHOW CREATE TABLE" command is not supported in Hive QL.

Answer: a) It displays the definition of a specified table, including the table schema and metadata.

Question:

In Hive QL, which statement is true about the "CROSS JOIN" operation?

- a) "CROSS JOIN" returns only the rows with matching values in both tables.
- b) It produces the Cartesian product of all rows from both tables.
- c) "CROSS JOIN" is used to combine rows from two tables based on a specified condition.
- d) The "CROSS JOIN" operation is not supported in Hive QL.



Answer: b) It produces the Cartesian product of all rows from both tables.

Question:

What does the following Hive QL query do?

SELECT department, COUNT(DISTINCT name) FROM employees GROUP BY department;

- a) Counts the total number of employees in each department.
- b) Retrieves the department names and the count of distinct employee names in each department.
- c) Retrieves the department names and the total number of employees in each department.
- d) Raises an error, as "DISTINCT" cannot be used with "COUNT" in Hive QL.

Answer: b) Retrieves the department names and the count of distinct employee names in each department.

Question:

Which Hive QL function is used to find the maximum value from a column?

- a) MAX()
- b) GREATEST()
- c) LARGEST()
- d) TOP()

Answer: a) MAX()

Question:

What is the result of the following Hive QL query?

SELECT concat('Hello', ' ', 'World') AS greeting;

- a) Retrieves the 'Hello World' string for each row in the table.
- b) Raises an error, as the "concat" function can only be used with numeric values.
- c) Returns the "greeting" column with the value 'Hello World'.
- d) Retrieves the values of columns named 'Hello' and 'World' from the table.

Answer: c) Returns the "greeting" column with the value 'Hello World'.

Question:

In Hive QL, what is the purpose of the "LOAD DATA" command?

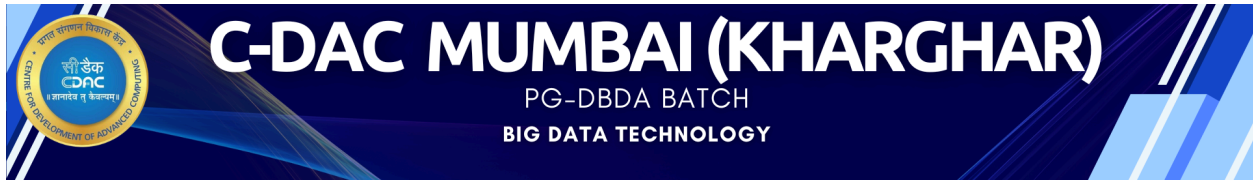
- a) It loads data from a local file system into a Hive table.
- b) The "LOAD DATA" command is used to insert data into a specific partition of a table.
- c) It is used to import data from an external database into Hive.
- d) The "LOAD DATA" command is not supported in Hive QL.

Answer: a) It loads data from a local file system into a Hive table.

Question:

Which Hive QL function is used to calculate the median of a numeric column?

- a) MEDIAN()
- b) MIDDLE()



c) PERCENTILE()

d) MIDDLE_VALUE()

Answer: c) PERCENTILE()

Question:

What is the purpose of the "SET" command in Hive QL?

a) It sets the value of a configuration parameter for the current session.

b) The "SET" command is used to define a new variable in Hive QL.

c) It sets the number of reducers to be used for a specific query.

d) The "SET" command is used to create and set the values of environment variables.

Answer: a) It sets the value of a configuration parameter for the current session.

Question:

In Hive QL, which statement is true about the "ORDER BY" clause in a SELECT query?

a) The "ORDER BY" clause is only used to sort data in ascending order.

b) It must always be the last clause in a SELECT query.

c) The "ORDER BY" clause is not allowed when using the "GROUP BY" clause in Hive QL.

d) It is used to sort the result set based on a specified column.

Answer: d) It is used to sort the result set based on a specified column

Question:

What is the purpose of the "SHOW FUNCTIONS" command in Hive QL?

a) It lists all the built-in functions available in Hive.

b) The "SHOW FUNCTIONS" command is used to view user-defined functions only.

c) It displays a list of all the functions used in a specific query.

d) The "SHOW FUNCTIONS" command is not supported in Hive QL.

Answer: a) It lists all the built-in functions available in Hive.

Question:

What does the following Hive QL query do?

sql

Copy code

```
SELECT department, SUM(salary) FROM employees GROUP BY department;
```

a) Retrieves the sum of salaries for each department.

b) Returns the total salary of all employees in each department.

c) Retrieves the department names and the sum of all employee salaries.

d) Raises an error, as "SUM()" cannot be used with "GROUP BY" in Hive QL.

Answer: a) Retrieves the sum of salaries for each department.

Question:

In Hive QL, which statement is true about the "LIKE" operator?



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- a) The "LIKE" operator is used to perform pattern matching on numeric columns.
- b) It supports regular expressions for pattern matching in Hive QL.
- c) The "LIKE" operator can only be used with string columns.
- d) The "LIKE" operator is not supported in Hive QL.

Answer: c) The "LIKE" operator can only be used with string columns.

Question:

What is the purpose of the "SHOW PARTITIONS" command in Hive QL?

- a) It lists all the partitions of a specific table.
- b) The "SHOW PARTITIONS" command is used to view the partitioning scheme of a table.
- c) It displays a list of all the available partition keys in Hive.
- d) The "SHOW PARTITIONS" command is not supported in Hive QL.

Answer: a) It lists all the partitions of a specific table.

Question:

Which Hive QL statement is used to change the data type of a column in an existing table?

- a) ALTER TABLE table_name CHANGE COLUMN column_name new_data_type;
- b) MODIFY COLUMN column_name new_data_type FOR table_name;
- c) ALTER TABLE table_name ALTER COLUMN column_name SET DATA TYPE new_data_type;
- d) UPDATE COLUMN column_name new_data_type IN table_name;

Answer: a) ALTER TABLE table_name CHANGE COLUMN column_name new_data_type;

Question:

In Hive QL, what is the purpose of the "AVG" function?

- a) It calculates the average value of a numeric column.
- b) The "AVG" function is used to find the most common value in a string column.
- c) It calculates the average of all the values in an array column.
- d) The "AVG" function returns the total number of rows in a result set.

Answer: a) It calculates the average value of a numeric column.

Question:

What is the purpose of the "GROUP BY" clause in a Hive QL query?

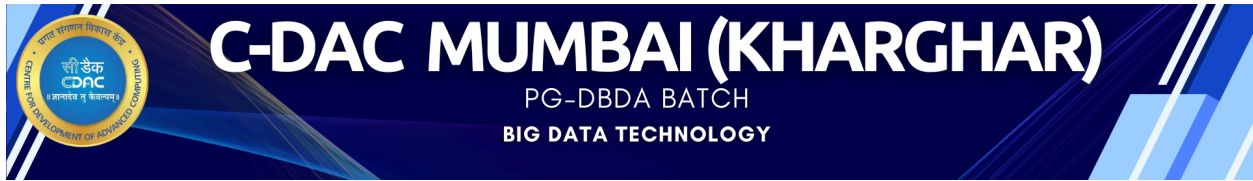
- a) It is used to combine multiple tables into a single result set.
- b) The "GROUP BY" clause sorts the result set based on a specified column.
- c) It groups the result set based on one or more columns, allowing aggregation functions to be applied.
- d) The "GROUP BY" clause is not allowed when using the "WHERE" clause in Hive QL.

Answer: c) It groups the result set based on one or more columns, allowing aggregation functions to be applied.

Question:

What does the following Hive QL query do?

sql



Copy code

```
SELECT name, salary FROM employees ORDER BY salary DESC;
```

- a) Retrieves the names and salaries of all employees, sorted in descending order of salary.
- b) Returns the names and salaries of employees, sorted alphabetically by name.
- c) Retrieves the highest salary and the corresponding employee's name.
- d) Raises an error, as "ORDER BY" cannot be used without an aggregate function.

Answer: a) Retrieves the names and salaries of all employees, sorted in descending order of salary.

Question:

In Hive QL, which statement is true about the "FROM" clause in a SELECT query?

- a) The "FROM" clause is used to specify the columns to be included in the query result.
- b) It defines the output format of the query result.
- c) The "FROM" clause is used to join multiple tables together.
- d) It is not required in a simple SELECT query.

Answer: c) The "FROM" clause is used to join multiple tables together.

Question:

What is the purpose of the "TRIM" function in Hive QL?

- a) It removes duplicates from the query result.
- b) The "TRIM" function is used to remove leading and trailing whitespace from a string.
- c) It removes NULL values from the query result.
- d) The "TRIM" function is used to truncate decimal places from numeric columns.

Answer: b) The "TRIM" function is used to remove leading and trailing whitespace from a string.

Question:

Which Hive QL function is used to calculate the total number of rows in a table?

- a) ROW_COUNT()
- b) COUNT_ROWS()
- c) ROW_NUMBER()
- d) COUNT()

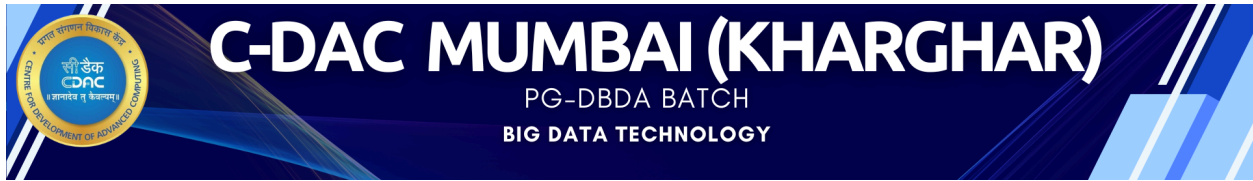
Answer: d) COUNT()

Question:

What is the purpose of the "JOIN" operation in Hive QL?

- a) The "JOIN" operation is used to combine rows from two or more tables based on a related column.
- b) It is used to merge data from two different databases.
- c) "JOIN" operation is used to sort the data in ascending order before applying the WHERE clause.
- d) The "JOIN" operation is not supported in Hive QL.

Answer: a) The "JOIN" operation is used to combine rows from two or more tables based on a related column.



Question:

In Hive QL, what will the following query do?

sql

Copy code

```
SELECT name, UPPER(department) FROM employees;
```

- a) Retrieves the names of employees along with the uppercase representation of the department names.
- b) Raises an error, as "UPPER" function can only be used with numeric columns.
- c) Returns the names of employees in uppercase letters.
- d) Retrieves the names of employees and the department names as they are in the "employees" table.

Answer: a) Retrieves the names of employees along with the uppercase representation of the department names.

Question:

What does the following Hive QL query do?

sql

Copy code

```
SELECT name, salary FROM employees WHERE salary > 50000;
```

- a) Retrieves the names and salaries of employees with a salary greater than \$50,000.
- b) Returns the names of employees who earn more than \$50,000.
- c) Retrieves the names and salaries of all employees, sorted by salary in descending order.
- d) Raises an error, as "WHERE" clause cannot be used with the "SELECT" statement.

Answer: a) Retrieves the names and salaries of employees with a salary greater than \$50,000.

Question:

Which Hive QL statement is used to create an external table?

- a) CREATE TABLE table_name AS SELECT * FROM source_table;
- b) CREATE EXTERNAL TABLE table_name (columns_definition) LOCATION 'hdfs_path';
- c) CREATE TABLE table_name (columns_definition) STORED AS ORC;
- d) CREATE EXTERNAL TABLE table_name LOCATION 'hdfs_path' AS SELECT * FROM source_table;

Answer: b) CREATE EXTERNAL TABLE table_name (columns_definition) LOCATION 'hdfs_path';

Question:

In Hive QL, what is the purpose of the "DISTRIBUTE BY" clause?

- a) The "DISTRIBUTE BY" clause is used to specify the number of reducers to be used for the query.
- b) It distributes the data across multiple nodes in the cluster based on a specified column.
- c) "DISTRIBUTE BY" is used to perform data distribution across all nodes in the cluster equally.
- d) The "DISTRIBUTE BY" clause is not supported in Hive QL.

Answer: b) It distributes the data across multiple nodes in the cluster based on a specified column.

Question:

What is the result of the following Hive QL query?

arduino

Copy code

```
SELECT CONCAT('Hello', ' ', 'World') AS greeting;
```

- a) Retrieves the 'Hello World' string for each row in the table.
- b) Raises an error, as the "CONCAT" function can only be used with numeric values.
- c) Returns the "greeting" column with the value 'Hello World'.
- d) Retrieves the values of columns named 'Hello' and 'World' from the table.

Answer: c) Returns the "greeting" column with the value 'Hello World'.

Question:

In Hive QL, which statement is true about the "ORC" file format?

- a) The "ORC" file format is the default file format used by Hive for storing tables.
- b) ORC stands for "Optimized Row Columnar," and it provides better compression and performance compared to other file formats.
- c) The "ORC" file format can only be used with external tables in Hive.
- d) ORC files are human-readable text files, making them easy to query directly.

Answer: b) ORC stands for "Optimized Row Columnar," and it provides better compression and performance compared to other file formats.

Question:

What does the following Hive QL query do?

sql

Copy code

```
SELECT COALESCE(name, 'Unknown') AS employee_name FROM employees;
```

- a) Retrieves the names of employees from the "employees" table.
- b) Returns the names of employees with any NULL values replaced by 'Unknown'.
- c) Retrieves the number of employees whose names are NULL.
- d) Raises an error, as "COALESCE" cannot be used without an aggregate function.

Answer: b) Returns the names of employees with any NULL values replaced by 'Unknown'.

Question:

Which Hive QL function is used to extract the year from a date column?

- a) DATEPART()
- b) EXTRACT_YEAR()
- c) YEAR()
- d) GET_YEAR()

Answer: c) YEAR()

Question:

What is the purpose of the Hive QL "SET hive.exec.dynamic.partition.mode" command?

- a) It sets the default value for dynamic partitioning in Hive QL.
- b) The "SET hive.exec.dynamic.partition.mode" command is used to enable dynamic partitioning for a specific query.
- c) It disables dynamic partitioning for the current Hive QL session.
- d) The "SET hive.exec.dynamic.partition.mode" command is not a valid Hive QL command.

Answer: a) It sets the default value for dynamic partitioning in Hive QL.

Question:

In Hive QL, which statement is true about the "ROUND" function?

- a) The "ROUND" function is used to calculate the sum of a numeric column.
- b) It rounds a numeric value to the nearest integer in Hive QL.
- c) The "ROUND" function is used to calculate the average of a numeric column.
- d) It truncates decimal places from a numeric column.

Answer: b) It rounds a numeric value to the nearest integer in Hive QL.

Question:

What is the purpose of the "INSERT INTO" statement in Hive QL?

- a) It inserts data into an existing table in Hive.
- b) The "INSERT INTO" statement is used to create a new table and insert data into it.
- c) It is used to join data from two or more tables into a new table.
- d) The "INSERT INTO" statement is not supported in Hive QL.

Answer: a) It inserts data into an existing table in Hive.

Question:

What does the following Hive QL query do?

sql

Copy code

```
SELECT DISTINCT department FROM employees;
```

- a) Retrieves all columns for distinct department names from the "employees" table.
- b) Returns the unique department names from the "employees" table.
- c) Retrieves the departments where there are no employees.
- d) Raises an error, as "DISTINCT" cannot be used without an aggregate function.

Answer: b) Returns the unique department names from the "employees" table.

Question:

In Hive QL, what will the following query do?

sql

Copy code

```
SELECT name, CONCAT('-', city, state) AS location FROM employees;
```

- a) Retrieves the names of employees along with their corresponding city and state separated by a hyphen.
- b) Raises an error, as "CONCAT_WS" function can only be used with numeric columns.
- c) Returns the "name" column with a hyphen concatenated between 'city' and 'state'.
- d) Retrieves the names of employees and the city and state columns as they are in the "employees" table.

Answer: a) Retrieves the names of employees along with their corresponding city and state separated by a hyphen.

Question:

What is MapReduce in the context of Hadoop?

- a) It is a programming language used to create data pipelines in Hadoop.
- b) MapReduce is a distributed storage system for large-scale data processing.
- c) It is a parallel processing technique used to analyze and process vast amounts of data.
- d) MapReduce is a NoSQL database for real-time data processing.

Answer: c) It is a parallel processing technique used to analyze and process vast amounts of data.

Question:

In MapReduce, what is the role of the Mapper function?

- a) The Mapper function is responsible for reducing the data before the final output.
- b) It is used to aggregate and summarize the data produced by the Reducer function.
- c) The Mapper function is responsible for processing input data and generating intermediate key-value pairs.
- d) The Mapper function is not used in the MapReduce framework.

Answer: c) The Mapper function is responsible for processing input data and generating intermediate key-value pairs.

Question:

What is the primary purpose of the Combiner function in MapReduce?

- a) The Combiner function is used to combine the output of multiple Mappers into a single file.
- b) It is used to optimize the data transfer between Mappers and Reducers.
- c) The Combiner function aggregates the output of multiple Mappers before sending it to the Reducer.
- d) The Combiner function is not a part of the MapReduce framework.

Answer: c) The Combiner function aggregates the output of multiple Mappers before sending it to the Reducer.

Question:

In MapReduce, what is the role of the Partitioner?

- a) The Partitioner is responsible for mapping each key-value pair to a specific Mapper.
- b) It is used to partition the data and distribute it across multiple nodes in the cluster.
- c) The Partitioner ensures that all keys with the same value are processed by the same Reducer.
- d) The Partitioner is not a part of the MapReduce framework.

Answer: c) The Partitioner ensures that all keys with the same value are processed by the same Reducer.

Question:

What does the "shuffle" phase in MapReduce refer to?

- a) It is the phase where the Mapper processes the input data and generates key-value pairs.
- b) The "shuffle" phase is responsible for distributing the intermediate data to the Reducers.
- c) It refers to the phase where the Reducers aggregate and summarize the intermediate data.
- d) The "shuffle" phase is not a part of the MapReduce process.

Answer: b) The "shuffle" phase is responsible for distributing the intermediate data to the Reducers.

Question:

What is the purpose of the InputFormat in MapReduce?

- a) The InputFormat defines how the input data is formatted before processing by the Mapper.
- b) It specifies the output format of the Reducer.
- c) The InputFormat is used to define the Mapper function.
- d) The InputFormat is not required in MapReduce.

Answer: a) The InputFormat defines how the input data is formatted before processing by the Mapper.

Question:

In MapReduce, what is the job of the Reducer function?

- a) The Reducer function is used to process input data and generate key-value pairs for the final output.
- b) It combines the output of multiple Mappers into a single output file.
- c) The Reducer function aggregates the intermediate key-value pairs and produces the final output.
- d) The Reducer function is responsible for sorting the output data.

Answer: c) The Reducer function aggregates the intermediate key-value pairs and produces the final output.

Question:

What is the purpose of the MapReduce jobtracker in Hadoop?

- a) The jobtracker is responsible for executing the Mapper and Reducer functions in parallel.
- b) It manages the metadata for the input and output data of MapReduce jobs.
- c) The jobtracker tracks the progress of MapReduce jobs and monitors cluster resources.
- d) The MapReduce jobtracker is not used in Hadoop.

Answer: c) The jobtracker tracks the progress of MapReduce jobs and monitors cluster resources.

Question:

In MapReduce, what is the "combinability" property?

- a) It refers to the ability of a MapReduce job to combine multiple input files into a single output file.
- b) The "combinability" property is used to control the number of Mappers and Reducers in a job.
- c) It is the property that allows the Reducer function to be used as a Combiner as well.
- d) The "combinability" property is not relevant in MapReduce.

Answer: c) It is the property that allows the Reducer function to be used as a Combiner as well.



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Question:

What is the purpose of the "cleanup" method in MapReduce?

- a) The "cleanup" method is used to delete temporary files generated during the MapReduce job.
- b) It is used to clean up the input data before processing by the Mapper.
- c) The "cleanup" method is responsible for cleaning up the output data before writing it to HDFS.
- d) The "cleanup" method is not a part of the MapReduce framework.

Answer: a) The "cleanup" method is used to delete temporary files generated during the MapReduce job.

Question:

What does the following Hadoop command do?

arduino

Copy code

```
hadoop fs -put local_file hdfs_path
```

- a) It puts the contents of the "local_file" on the Hadoop Distributed File System (HDFS) at the specified "hdfs_path."
- b) The command retrieves the contents of the "local_file" from HDFS and saves it to the "hdfs_path" on the local file system.
- c) It deletes the "local_file" from the local file system.
- d) The command creates a new directory with the name "local_file" in HDFS.

Answer: a) It puts the contents of the "local_file" on the Hadoop Distributed File System (HDFS) at the specified "hdfs_path."

Question:

What is the purpose of the "setup" method in MapReduce?

- a) The "setup" method is used to initialize variables and resources before the Mapper or Reducer functions start processing.
- b) It is used to set up the jobtracker and tasktracker for MapReduce jobs.
- c) The "setup" method is responsible for loading input data into the MapReduce job.
- d) The "setup" method is not a part of the MapReduce framework.

Answer: a) The "setup" method is used to initialize variables and resources before the Mapper or Reducer functions start processing.

Question:

In MapReduce, what does the "mapred.reduce.tasks" property define?

- a) The "mapred.reduce.tasks" property defines the number of Reducers to be used for a specific job.
- b) It specifies the size of the input data for MapReduce jobs.
- c) The "mapred.reduce.tasks" property is used to enable or disable the use of the Combiner function.
- d) The property "mapred.reduce.tasks" is not relevant in MapReduce.

Answer: a) The "mapred.reduce.tasks" property defines the number of Reducers to be used for a specific job.

Question:

What is the output key type in MapReduce?

- a) The output key type is determined by the InputFormat used in the MapReduce job.
- b) It is the type of the key produced by the Mapper and consumed by the Reducer.
- c) The output key type is specified in the Hadoop configuration files.
- d) The output key type is not relevant in MapReduce.

Answer: b) It is the type of the key produced by the Mapper and consumed by the Reducer.

Question:

What is the purpose of the "combine" method in MapReduce?

- a) The "combine" method is used to merge the output of multiple Reducers before writing it to HDFS.
- b) It is used to combine the intermediate key-value pairs produced by the Mapper function.
- c) The "combine" method is responsible for performing a preliminary aggregation on the Mapper output.
- d) The "combine" method is not a part of the MapReduce framework.

Answer: c) The "combine" method is responsible for performing a preliminary aggregation on the Mapper output.

Question:

What does the following Hadoop command do?

bash

Copy code

```
hadoop fs -ls hdfs_path
```

- a) It lists all the files and directories in the Hadoop Distributed File System (HDFS) at the specified "hdfs_path."
- b) The command lists all the files and directories in the current local file system.
- c) It creates a new directory with the name "hdfs_path" in HDFS.
- d) The command displays the properties and configurations of HDFS.

Answer: a) It lists all the files and directories in the Hadoop Distributed File System (HDFS) at the specified "hdfs_path."

Question:

In MapReduce, what is the purpose of the "InputSplit" class?

- a) The "InputSplit" class is used to define the input data for the MapReduce job.
- b) It specifies the number of Mappers and Reducers for a particular job.
- c) The "InputSplit" class is responsible for dividing the input data into manageable chunks for processing.
- d) The "InputSplit" class is not used in the MapReduce framework.

Answer: c) The "InputSplit" class is responsible for dividing the input data into manageable chunks for processing.

Question:



What is the purpose of the "IdentityReducer" class in MapReduce?

- a) The "IdentityReducer" class is used when no Reducer logic is required for a MapReduce job.
- b) It is used to combine the output of multiple Mappers before sending it to the Reducer.
- c) The "IdentityReducer" class is responsible for sorting the output data.
- d) The "IdentityReducer" class is not relevant in MapReduce.

Answer: a) The "IdentityReducer" class is used when no Reducer logic is required for a MapReduce job.

Question:

What does the following Hadoop command do?

Copy code

```
hadoop jar myJob.jar com.example.MyJob input_dir output_dir
```

- a) It runs the "MyJob" MapReduce job on the input data from "input_dir" and stores the result in "output_dir."
- b) The command compiles the "MyJob" class and creates a JAR file named "myJob.jar."
- c) It lists all the MapReduce jobs submitted by the user.
- d) The command generates a report on the status of the Hadoop cluster.

Answer: a) It runs the "MyJob" MapReduce job on the input data from "input_dir" and stores the result in "output_dir."

Question:

What is the purpose of the "FileInputFormat" in MapReduce?

- a) The "FileInputFormat" specifies the format of the output data produced by the Mapper.
- b) It is used to define the format of the input data before processing by the Mapper.
- c) The "FileInputFormat" is responsible for setting up the jobtracker and tasktracker for MapReduce jobs.
- d) The "FileInputFormat" is not relevant in MapReduce.

Answer: b) It is used to define the format of the input data before processing by the Mapper.

Question:

What does the following Hadoop command do?

bash

Copy code

```
hadoop fs -cat hdfs_path
```

- a) It displays the content of the "hdfs_path" on the local file system.
- b) The command concatenates the contents of all files in "hdfs_path" and displays the output.
- c) It creates a new file with the name "hdfs_path" in HDFS.
- d) The command copies the "hdfs_path" to a different location in HDFS.

Answer: b) The command concatenates the contents of all files in "hdfs_path" and displays the output.

Question:

In MapReduce, what is the purpose of the "TextInputFormat"?



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- a) The "TextInputFormat" specifies the format of the input data for Reducers.
- b) It is used to define the format of the input data for Mappers.
- c) The "TextInputFormat" is responsible for converting text-based input data into key-value pairs.
- d) The "TextInputFormat" is not used in the MapReduce framework.

Answer: c) The "TextInputFormat" is responsible for converting text-based input data into key-value pairs.

Question:

What does the following Hadoop command do?

mathematica

Copy code

```
hadoop jar myJob.jar com.example.MyJob -D mapred.reduce.tasks=5 input_dir output_dir
```

- a) It runs the "MyJob" MapReduce job on the input data from "input_dir" and stores the result in "output_dir" with five Reducers.
- b) The command compiles the "MyJob" class and creates a JAR file named "myJob.jar" with five Reducers specified.
- c) It lists all the MapReduce jobs submitted by the user with the option to specify the number of Reducers.
- d) The command generates a report on the status of the Hadoop cluster with five Reducers.

Answer: a) It runs the "MyJob" MapReduce job on the input data from "input_dir" and stores the result in "output_dir" with five Reducers specified.

Question:

What is the purpose of the "SequenceFileInputFormat" in MapReduce?

- a) The "SequenceFileInputFormat" specifies the format of the output data produced by the Reducer.
- b) It is used to define the format of the input data before processing by the Reducer.
- c) The "SequenceFileInputFormat" is responsible for converting sequence files into key-value pairs.
- d) The "SequenceFileInputFormat" is not relevant in MapReduce.

Answer: c) The "SequenceFileInputFormat" is responsible for converting sequence files into key-value pairs.

Question:

In MapReduce, what is the purpose of the "Writable" interface?

- a) The "Writable" interface is used to define the format of the input data for Reducers.
- b) It specifies the format of the output data produced by the Mapper.
- c) The "Writable" interface is responsible for serializing and deserializing data in MapReduce jobs.
- d) The "Writable" interface is not used in the MapReduce framework.

Answer: c) The "Writable" interface is responsible for serializing and deserializing data in MapReduce jobs.



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What is the primary purpose of ETL in Hadoop?

- a) To load data into Hadoop from various sources.
- b) To process and analyze data in Hadoop.
- c) To extract data from Hadoop and load it into a data warehouse.
- d) To transform and cleanse data before storing it in Hadoop.

Answer: d) To transform and cleanse data before storing it in Hadoop.

Which Hadoop component is responsible for data loading?

- a) HDFS
 - b) YARN
 - c) Hive
 - d) Sqoop
- Answer: d) Sqoop

What does Sqoop stand for in Hadoop ETL?

- a) SQL Query Optimization
- b) Structured Query Operations
- c) SQL-to-Hadoop OPerations
- d) Structured Query Optimizer

Answer: c) SQL-to-Hadoop OPerations

Which Hadoop tool is best suited for importing data from a relational database into Hadoop?

- a) Flume
 - b) Pig
 - c) Sqoop
 - d) Hive
- Answer: c) Sqoop

In Hadoop ETL, what is the purpose of the "mapper" in MapReduce?

- a) To read data from HDFS and send it to the reducer.
- b) To transform and cleanse data before loading it into HDFS.
- c) To process data in parallel across the cluster.
- d) To load data from external sources into Hadoop.

Answer: c) To process data in parallel across the cluster.

Which of the following is a data serialization format commonly used in Hadoop ETL?

- a) JSON
- b) XML
- c) YAML
- d) All of the above

Answer: d) All of the above



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In Hadoop ETL, what is the primary purpose of the "reducer" in MapReduce?

- a) To read data from HDFS and send it to the mapper.
- b) To combine and aggregate data from the mapper phase.
- c) To transform data from one format to another.
- d) To optimize SQL queries in Hadoop.

Answer: b) To combine and aggregate data from the mapper phase.

Which Hadoop tool is used for data ingestion from log files and streaming data sources?

- a) Pig
- b) Hive
- c) Flume
- d) Sqoop

Answer: c) Flume

What is the advantage of using Apache Pig in Hadoop ETL?

- a) It provides a graphical user interface for data transformation.
- b) It supports complex data transformations without writing Java code.
- c) It is the fastest tool for loading data into Hadoop.
- d) It provides real-time data processing capabilities.

Answer: b) It supports complex data transformations without writing Java code.

Which of the following is NOT a valid data source for Hadoop ETL using Flume?

- a) Apache Kafka
- b) Web Server logs
- c) Relational databases
- d) Twitter feeds

Answer: c) Relational databases

In Hadoop ETL, what is the function of the "loader" component?

- a) To load data into Hadoop from various sources.
- b) To schedule and coordinate data processing tasks.
- c) To transform and cleanse data before storing it in HDFS.
- d) To transfer data between Hadoop and external systems.

Answer: a) To load data into Hadoop from various sources.

Which Hadoop tool allows you to query and analyze data using SQL-like syntax?

- a) Pig
- b) Hive
- c) Sqoop
- d) Flume

Answer: b) Hive

What is the benefit of using partitioning in Hadoop ETL?

- a) It allows for data replication across multiple nodes for fault tolerance.
- b) It optimizes data storage and retrieval by dividing data into manageable segments.
- c) It enables real-time data streaming and processing.
- d) It simplifies the process of data transformation.

Answer: b) It optimizes data storage and retrieval by dividing data into manageable segments.

What is the purpose of the "schema" in Hive for Hadoop ETL?

- a) It determines the order of data processing in MapReduce.
- b) It defines the structure of the data stored in HDFS.
- c) It specifies the format and location of the data in HDFS.
- d) It provides a logical view of the data for querying.

Answer: d) It provides a logical view of the data for querying.

Which Hadoop tool is best suited for batch processing of large datasets?

- a) Pig
- b) Hive
- c) Flume
- d) Sqoop

Answer: a) Pig

What is the primary function of the "data cleaning" phase in Hadoop ETL?

- a) To load data into Hadoop from various sources.
- b) To optimize SQL queries in Hadoop.
- c) To transform and cleanse data before storing it in HDFS.
- d) To process data in parallel across the cluster.

Answer: c) To transform and cleanse data before storing it in HDFS.

Which Hadoop tool is used for data preprocessing and data transformation?

- a) Pig
- b) Hive
- c) Flume
- d) Sqoop

Answer: a) Pig

What is the advantage of using Hadoop for ETL compared to traditional ETL tools?

- a) Hadoop provides real-time data processing capabilities.
- b) Hadoop allows for cost-effective storage and processing of large volumes of data.
- c) Hadoop ETL tools have a graphical user interface for easy data transformation.
- d) Hadoop eliminates the need for data serialization and deserialization.

Answer: b) Hadoop allows for cost-effective storage and processing of large volumes of data.

Which of the following is NOT a core component of Hadoop?

- a) HDFS
- b) MapReduce
- c) YARN
- d) Hive

Answer: d) Hive

In Hadoop ETL, what is the purpose of the "shuffle and sort" phase in MapReduce?

- a) To load data from external sources into Hadoop.
- b) To transform and cleanse data before loading it into HDFS.
- c) To group and sort the data before sending it to the reducers.
- d) To schedule and coordinate data processing tasks.

Answer: c) To group and sort the data before sending it to the reducers.

Which Hadoop tool is used for data warehousing and SQL-based querying of data?

- a) Pig
- b) Hive
- c) Flume
- d) Sqoop

Answer: b) Hive

What is the primary function of the "reduce" function in MapReduce?

- a) To read data from HDFS and send it to the mapper.
- b) To combine and aggregate data from the mapper phase.
- c) To optimize SQL queries in Hadoop.
- d) To process data in parallel across the cluster.

Answer: b) To combine and aggregate data from the mapper phase.

In Hadoop ETL, what is the role of the "source connector" in Flume?

- a) To read data from external sources and send it to Hadoop.
- b) To process data in parallel across the cluster.
- c) To transfer data between Hadoop and external systems.
- d) To load data into Hadoop from various sources.

Answer: a) To read data from external sources and send it to Hadoop.

Which of the following is NOT a supported data format in Hadoop ETL?

- a) Avro
- b) Parquet
- c) ORC
- d) CSV

Answer: d) CSV

What is the primary benefit of using Apache NiFi in Hadoop ETL?

- a) It provides a graphical user interface for data transformation.
- b) It supports complex data transformations without writing Java code.
- c) It allows real-time data processing and streaming.
- d) It is the fastest tool for loading data into Hadoop.

Answer: c) It allows real-time data processing and streaming.

Which Hadoop tool is used for data flow management and data routing?

- a) Pig
- b) Hive
- c) Flume
- d) Sqoop

Answer: c) Flume

What is the primary advantage of using HCatalog in Hadoop ETL?

- a) It provides a centralized metadata repository for all Hadoop components.
- b) It supports complex data transformations using a SQL-like language.
- c) It allows for real-time data streaming and processing.
- d) It provides a graphical user interface for data transformation.

Answer: a) It provides a centralized metadata repository for all Hadoop components.

In Hadoop ETL, what is the role of the "sink connector" in Flume?

- a) To read data from external sources and send it to Hadoop.
- b) To process data in parallel across the cluster.
- c) To transfer data between Hadoop and external systems.
- d) To load data into Hadoop from various sources.

Answer: c) To transfer data between Hadoop and external systems.

Which Hadoop tool is best suited for processing streaming data in real-time?

- a) Pig
- b) Hive
- c) Flume
- d) Spark Streaming

Answer: d) Spark Streaming

What is the primary purpose of the "split" function in Hadoop ETL?

- a) To transform and cleanse data before loading it into HDFS.
- b) To load data from external sources into Hadoop.
- c) To schedule and coordinate data processing tasks.
- d) To divide input data into smaller chunks for parallel processing.

Answer: d) To divide input data into smaller chunks for parallel processing.



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Which Hadoop tool is used for data storage and retrieval in a columnar format?

- a) Pig
- b) Hive
- c) HBase
- d) Sqoop

Answer: c) HBase

What is the primary purpose of the "combiner" in Hadoop ETL?

- a) To read data from HDFS and send it to the mapper.
- b) To combine and aggregate data from the mapper phase.
- c) To transform and cleanse data before loading it into HDFS.
- d) To process data in parallel across the cluster.

Answer: b) To combine and aggregate data from the mapper phase.

Which Hadoop tool is best suited for data processing in a near-real-time fashion?

- a) Pig
- b) Hive
- c) Flume
- d) Apache Storm

Answer: d) Apache Storm

What is the purpose of the "data enrichment" phase in Hadoop ETL?

- a) To transform and cleanse data before loading it into HDFS.
- b) To optimize SQL queries in Hadoop.
- c) To load data into Hadoop from various sources.
- d) To enhance data with additional information from external sources.

Answer: d) To enhance data with additional information from external sources.

Which of the following is NOT a valid data storage format for Hadoop ETL?

- a) JSON
- b) ORC
- c) XML
- d) SQL

Answer: d) SQL

What is the primary advantage of using Apache Kafka in Hadoop ETL?

- a) It provides a graphical user interface for data transformation.
- b) It supports complex data transformations using a SQL-like language.
- c) It allows for real-time data streaming and processing.
- d) It is the fastest tool for loading data into Hadoop.

Answer: c) It allows for real-time data streaming and processing.

Which Hadoop tool is used for data serialization and deserialization?

- a) Pig
- b) Hive
- c) Avro
- d) Sqoop

Answer: c) Avro

What is the primary purpose of the "transformation" phase in Hadoop ETL?

- a) To load data into Hadoop from various sources.
- b) To process data in parallel across the cluster.
- c) To transform and cleanse data before loading it into HDFS.
- d) To schedule and coordinate data processing tasks.

Answer: c) To transform and cleanse data before loading it into HDFS.

Which Hadoop tool is best suited for data streaming and complex event processing?

- a) Pig
- b) Hive
- c) Flume
- d) Apache Flink

Answer: d) Apache Flink

What is the primary purpose of the "join" operation in Hadoop ETL?

- a) To combine and aggregate data from multiple sources.
- b) To optimize SQL queries in Hadoop.
- c) To transform and cleanse data before loading it into HDFS.
- d) To merge data from different datasets based on a common key.

Answer: d) To merge data from different datasets based on a common key.

Which Hadoop tool is used for data ingestion from social media platforms?

- a) Pig
- b) Hive
- c) Flume
- d) Sqoop

Answer: c) Flume

What is the purpose of the "data loading" phase in Hadoop ETL?

- a) To transform and cleanse data before loading it into HDFS.
- b) To load data into Hadoop from various sources.
- c) To optimize SQL queries in Hadoop.
- d) To process data in parallel across the cluster.

Answer: b) To load data into Hadoop from various sources.



Which Hadoop tool is used for data processing using a procedural language called Pig Latin?

- a) Pig
- b) Hive
- c) Flume
- d) Sqoop

Answer: a) Pig

What is the primary advantage of using Apache NiFi in Hadoop ETL?

- a) It provides a graphical user interface for data transformation.
- b) It supports complex data transformations without writing Java code.
- c) It allows real-time data processing and streaming.
- d) It is the fastest tool for loading data into Hadoop.

Answer: c) It allows real-time data processing and streaming.

Which of the following is NOT a core component of Hadoop?

- a) HDFS
- b) MapReduce
- c) YARN
- d) Hive

Answer: d) Hive

In Hadoop ETL, what is the purpose of the "shuffle and sort" phase in MapReduce?

- a) To load data from external sources into Hadoop.
- b) To transform and cleanse data before loading it into HDFS.
- c) To group and sort the data before sending it to the reducers.
- d) To schedule and coordinate data processing tasks.

Answer: c) To group and sort the data before sending it to the reducers.

Which Hadoop tool is used for data warehousing and SQL-based querying of data?

- a) Pig
- b) Hive
- c) Flume
- d) Sqoop

Answer: b) Hive

What is the primary function of the "reduce" function in MapReduce?

- a) To read data from HDFS and send it to the mapper.
- b) To combine and aggregate data from the mapper phase.
- c) To optimize SQL queries in Hadoop.
- d) To process data in parallel across the cluster.

Answer: b) To combine and aggregate data from the mapper phase.



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BIG DATA TECHNOLOGY

In Hadoop ETL, what is the role of the "source connector" in Flume?

- a) To read data from external sources and send it to Hadoop.
- b) To process data in parallel across the cluster.
- c) To transfer data between Hadoop and external systems.
- d) To load data into Hadoop from various sources.

Answer: a) To read data from external sources and send it to Hadoop.

Which of the following is NOT a supported data format in Hadoop ETL?

- a) Avro
- b) Parquet
- c) ORC
- d) CSV

Answer: d) CSV